

CHRONO-Fair: A Controlled Validation Study of Anytime-Valid Counterfactual Fairness Monitoring for Clinical Machine Learning

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The use of machine learning in clinical care is expanding rapidly, and with it the catalogue of subgroup-specific fairness failures that surface only after release. The auditing tools in routine use respond to this by computing a group fairness metric on a fixed snapshot of predictions and returning a pass-or-fail verdict against a threshold such as the 0.8 rule. However, a point-estimate verdict hides whether the underlying number is stable: on the real Texas-100X discharge dataset (925 128 records), the worst-group true-positive-rate verdict for race reads “fair” at 0.824, yet the same verdict flips to “unfair” in 4 of 20 hospital-network resamples, a Verdict Flip Rate of 0.20. No batch audit will see when, or whether, that flip recurs as new patients arrive.

This paper presents CHRONO-Fair, a prototype monitoring framework evaluated as a controlled validation study. The main contribution is a per-cell anytime-valid e-process on a counterfactual flip indicator, with an inspection-time cross-cell adjustment. Three supporting modules accompany it: a Flip Hazard survival view of group-stratified time-to-flip, an entropy-based diagnostic triage that summarises whether a flip signal is more consistent with data-driven noise, model uncertainty, or a mixed pattern, and a Wasserstein-1 rank-allocation statistic for length-of-stay regression. The evidence base combines a real Texas-100X batch verdict-stability analysis with Texas-100X-calibrated controlled streaming and retrospective replay experiments. On the streams, the e-process detected an injected drift in 50 of 50 Monte-Carlo runs at a median delay of 828 patients, against 21 of 50 for ADWIN, with the empirical false-alarm rate at or below the nominal level for every tested α . The streaming evidence is controlled replay and simulation rather than prospective clinical validation, and prospective evaluation on a live electronic-health-record stream is the necessary next step before any operational adoption.

Additional Key Words and Phrases: algorithmic fairness, clinical machine learning, anytime-valid inference, fairness drift, post-deployment monitoring, length-of-stay prediction

1 INTRODUCTION

Machine learning is now embedded across many parts of clinical practice, from sepsis early warning to length-of-stay forecasting and discharge planning. The catalogue of documented fairness failures has expanded in step. Obermeyer et al. [82] traced racial bias in a widely used care-management algorithm to its cost-as-proxy label, and reported that correcting the label nearly tripled the share of Black patients flagged for additional care. Seyyed-Kalantari et al. [98] reported higher underdiagnosis rates for female and minority patients across multi-site chest-radiograph classifiers, with the largest gaps on intersectional subgroups. Habib et al. [44] and Wong et al. [114] found that the Epic Sepsis Model achieved an external-validation area under the receiver operating characteristic curve of 0.63, well below the vendor-reported range. Davis et al. [29] extended the picture along the time axis, showing across eleven years of VA cohort data that fairness gaps drift after release, often toward widening.

These post-release failures have pushed regulatory and reporting expectations from one-off model validation toward continuous lifecycle monitoring. The U.S. FDA codified the shift in its December 2024 Predetermined Change Control Plan final guidance [103]; Article 17 of the EU AI Act, binding from August 2027 for high-risk medical AI [37], requires continuous post-market bias monitoring under a documented plan; and the DECIDE-AI [105], TRIPOD+AI [26], and CONSORT-AI [69] reporting standards treat lifecycle evaluation as expected practice rather than as an extension. The clinical-informatics literature has begun

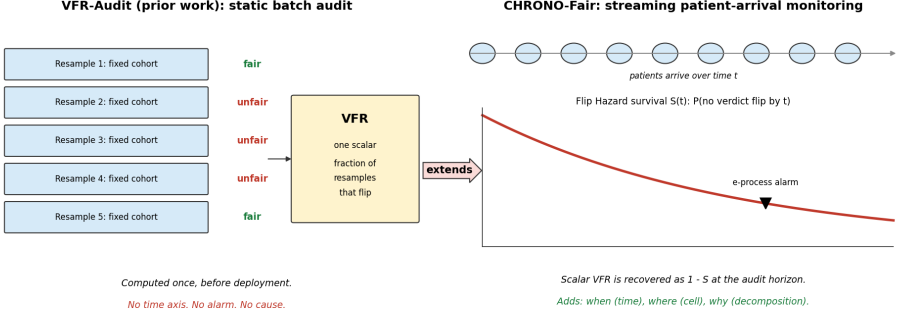


Fig. 1. VFR-Audit is a static batch scalar. CHRONO-Fair is the streaming extension. The scalar Verdict Flip Rate is recovered as $1 - S$ at the audit horizon, and the framework adds the time axis, the alarm, and the diagnostic triage.

to operationalise the same shift: You et al. [115] propose a clinical-trials-informed four-phase framework that places post-deployment monitoring as a co-equal phase with safety, efficacy, and effectiveness; Ansari et al. [8] survey the open challenges of postmarket surveillance for clinical prediction models. McCradden et al. [75] and Eaneff et al. [36] argue, in the same direction, for “algorithmic stewardship” as a regulatory category.

The toolchain available to satisfy that expectation, however, has not caught up. Aequitas, Fairlearn, AIF360, and the Microsoft Responsible AI Dashboard, the auditing packages in routine use, all operate in batch mode on a fixed set of predictions. Online fair-learning work [50, 51, 81] monitors a stream but has been demonstrated on COMPAS and credit data rather than on a clinical cohort. Anytime-valid inference [43, 48, 92, 113] supplies the sequential statistical machinery, yet it has been applied to A/B-test monitoring and calibration drift rather than to fairness drift at intersectional granularity. Predictive-multiplicity work [13, 27, 73] characterises pipeline instability but frames it as reliability rather than fairness. None of these lines reports when a thresholded fairness verdict, the object hospitals actually act on, begins to flip as patients arrive.

The closest precedent on the verdict-stability question is the prior VFR-Audit work [7], which introduced the Verdict Flip Rate, a scalar that quantifies how often a fairness verdict flips between resamples of a hospital-network bootstrap. It captures something the canonical group metrics miss, the instability of the verdict itself, and it does so on the pass-or-fail object that governance teams act on. It does so, however, as one batch number at one threshold, and it is silent on when, where, or why a verdict moves after release. The present paper extends that line along the time axis.

Contribution. The framework introduced in this paper, CHRONO-Fair, sits above a deployed clinical machine learning model and monitors whether its thresholded fairness verdict continues to hold as patients arrive. The deployed model may be uncorrected or may have been fairness-adjusted by any pre-release mitigation method; CHRONO-Fair is agnostic to that choice. This paper makes four contributions, evaluated through real Texas-100X batch evidence and Texas-100X-calibrated controlled streaming and replay experiments. The main contribution is a per-cell e-process on a counterfactual flip indicator, calibrated against a pre-deployment baseline flip rate and combined with an inspection-time

cross-cell adjustment. Three supporting modules give the governance-review interface its evidence: a Flip Hazard survival view of group-stratified time-to-flip, an entropy-based diagnostic attribution that summarises whether a detected flip signal is more consistent with data-driven noise, model uncertainty, or a mixed pattern, and a Wasserstein-1 rank-allocation statistic for length-of-stay regression. A dashboard and an automatically generated inspection report bundle the four outputs for a clinical safety officer. The novelty is not a new survival estimator or a new e-process. The contribution is the integration of time-resolved flip-hazard estimation, anytime-valid per-cell alarm evidence, diagnostic flip attribution, and rank-allocation regression monitoring around the governance object that hospitals act on: the thresholded fairness verdict.

2 RELATED WORK

2.1 Group, individual, and counterfactual fairness

Dwork et al. [35] defined individual fairness as a Lipschitz constraint. Two individuals within distance d on a task metric must receive outcome distributions within distance D . Hardt et al. [45] defined equalised odds as equal true-positive and false-positive rates across groups. Equal opportunity relaxes this to equal true-positive rate only. Chouldechova [24] proved that, when group base rates differ, a classifier cannot hold predictive parity, equal false-positive rate, and equal false-negative rate at once. Kleinberg et al. [63] proved a parallel result: calibration and class-balanced error are jointly attainable only under equal base rates or perfect prediction. Pleiss et al. [87] showed the conflict persists for any non-trivial calibration constraint and quantified the achievable relaxation. Kusner et al. [65] defined counterfactual fairness as invariance of the prediction distribution under a do-intervention on the protected attribute in a structural causal model. Chiappa [23] restricted this to unfair causal paths only, leaving resolving paths intact. Nabi and Shpitser [78] constrained path-specific effects during estimation by a constrained likelihood. Kilbertus et al. [61] separated proxy discrimination from explanatory effects in a causal graph. Makhoul et al. [71] mapped 19 causal fairness notions to the data assumptions each one needs. The surveys of Verma and Rubin [106], Mehrabi et al. [76], Pessach and Shmueli [85], and Caton and Haas [19] catalogue 20 or more metrics. They report that the choice of metric, not the model, drives most reported disparity. On mitigation, Agarwal et al. [2] reduced fair classification to a sequence of cost-weighted subproblems solved by a saddle-point game. Donini et al. [32] added a fairness term to empirical risk minimisation with a generalisation bound. Calmon et al. [17] and Kamiran and Calders [57] reweighted or transformed training data before fitting. Madras et al. [70] and Lahoti et al. [66] learned representations that an adversary cannot use to recover the protected attribute. Wang et al. [111] bounded fairness violation when group labels are noisy. Kallus et al. [56] estimated disparity by combining a main dataset with an auxiliary dataset that carries the protected attribute.

2.2 Subgroup, multicalibration, and statistical-audit perspectives

Kearns et al. [59] showed that parity on coarse groups can hide large violations on structured subgroups, an effect they named fairness gerrymandering. Hébert-Johnson et al. [47] defined multicalibration as calibration that holds simultaneously on every subgroup in a computationally identifiable class. Kim et al. [62] produced multiaccuracy by post-processing, auditing for subgroups where the residual correlates with subgroup membership. Hashimoto et al. [46], Sagawa et al. [94], and Duchi and Namkoong [33] minimised worst-group loss through distributionally robust optimisation over an uncertainty set of group mixtures. Martínez et

al. [72] characterised the accuracy-fairness frontier as a Pareto set and computed it through a multi-objective descent. Cherian and Candès [22] produced finite-sample confidence intervals for a fairness metric using bootstrap and rank-based statistics. Besse et al. [11] reported that the sampling variance of statistical parity can exceed the disparity itself when subgroup size is below a few hundred. Coston et al. [28] evaluated counterfactual risk under selective labels through inverse-propensity weighting. These works audit a fixed snapshot. None updates a fairness verdict sequentially as patients arrive.

2.3 Predictive multiplicity, model multiplicity, and stability

Breiman [16] observed that distinct models can attain near-identical loss while disagreeing on individuals. Marx et al. [73] formalised this as predictive multiplicity. They defined ambiguity as the fraction of inputs that receive a conflicting prediction from some model within an ϵ -loss Rashomon set. Watson-Daniels et al. [112] extended the measure to probabilistic outputs through the range of predicted scores. Semenova et al. [97] linked a large Rashomon set to the existence of a simpler accurate model. Dong and Rudin [31] computed variable-importance ranges across the Rashomon set rather than for one model. Black et al. [13] argued that model multiplicity gives a designer latitude to select a fairer model at no accuracy cost. Cooper et al. [27] showed that a measured group gap can be a variance artefact: averaging predictions across the Rashomon set shrank reported disparity. Miller and Blume [77] defined the empirical Decision Flip Rate as the fraction of patients whose binary decision changes between bootstrap re-trainings. Black et al. [14] detected individual discrimination through an optimal-transport map between group-conditional input distributions. The Verdict Flip Rate of [7] measured how often the fairness verdict, not the prediction, flips between hospital-network resamples.

2.4 Uncertainty quantification and its decomposition

Depeweg et al. [30] decomposed predictive uncertainty into two terms. Aleatoric uncertainty is the mean of per-model entropies. Epistemic uncertainty is the entropy of the mean prediction minus that mean, equal to the mutual information between label and parameters. Hüllermeier and Waegeman [49] formalised the distinction and noted that only the epistemic term shrinks with more data. Lakshminarayanan et al. [67] estimated epistemic uncertainty from the disagreement of M independently trained networks. Gal and Ghahramani [40] approximated the same quantity through Monte Carlo dropout. Kendall and Gal [60] separated learned input-dependent variance from model uncertainty in a single loss. Ovadia et al. [83] reported that deep ensembles kept calibration error lowest among uncertainty methods under distribution shift. Alghamdi et al. [4] applied the aleatoric-epistemic split to discrimination, attributing the irreducible part to the data and the reducible part to the model class. Romano et al. [93] and Gibbs and Candès [42] produced prediction sets with a finite-sample coverage guarantee, the latter adapting the set width online under shift. Barber et al. [9] extended conformal coverage to non-exchangeable data with a bound that degrades in the total-variation distance from exchangeability. Angelopoulos and Bates [6] surveyed the coverage guarantees that the conformal family provides. Dutta et al. [34] and Calmon et al. [18] decomposed discrimination through partial information, separating the exempt component from the unfair component.

2.5 Anytime-valid inference, drift detection, online learning

Ville [107] proved that a non-negative supermartingale started at one crosses $1/\alpha$ with probability at most α . This single inequality underwrites every e-process. Wald [110] built

the sequential probability ratio test, which stops as soon as a likelihood ratio leaves a fixed interval. Howard et al. [48] produced confidence sequences, intervals that cover the parameter simultaneously at every sample size with probability $1 - \alpha$. Waudby-Smith and Ramdas [113] estimated a bounded mean by betting. The wealth process $\prod_t (1 + \lambda_t (X_t - m))$ is a martingale under the null mean m . The GROW rule picks λ_t to maximise expected log-growth. Ramdas et al. [92] and Shafer [99] unified these results as game-theoretic statistics, where evidence equals the capital won against the null. Grünwald et al. [43] characterised the e-variable that is growth-optimal against a composite alternative. Johari et al. [54] applied always-valid p -values to continuous A/B-test monitoring, removing the penalty for repeated peeking. Podkopaev and Ramdas [88] tracked a deployed model’s risk through a betting martingale on the loss sequence. Foster and Stine [39] introduced alpha-investing, which spends a false-discovery budget across an ordered hypothesis stream. Javanmard and Montanari [52] produced the LORD rules that control online false-discovery rate with a decaying memory. Bifet and Gavalda [12] introduced ADWIN, which keeps a variable-length window and drops the older half when two sub-window means differ by a Hoeffding bound. Gama et al. [41] taxonomised concept drift into sudden, gradual, and recurring types. Rabanser et al. [90] reported that dimensionality reduction followed by a two-sample test detected dataset shift most reliably. Iosifidis and Ntoutsis [50, 51] maintained a fairness constraint online by reweighting the loss as a stream evolves. None of these monitors a counterfactual flip indicator under intersectional multiple-testing control.

2.6 Clinical machine learning fairness, prediction, and deployment

Obermeyer et al. [82] traced the bias of a care-management algorithm to its label. The model predicted healthcare cost as a proxy for healthcare need. At a fixed risk score, Black patients carried more chronic conditions than White patients. Correcting the label raised the share of Black patients flagged for extra care from 17.7% to 46.5%. Vyas et al. [108] catalogued race-adjusted clinical formulas, including an eGFR correction that lowered estimated kidney disease severity for Black patients. Seyyed-Kalantari et al. [98] reported that chest radiograph classifiers had higher false-negative rates for female, Black, and publicly insured patients, with the largest underdiagnosis on intersectional subgroups. Wong et al. [114] externally validated the Epic Sepsis Model. They found an area under the receiver operating characteristic curve of 0.63, against the vendor-reported 0.76 to 0.83. Habib et al. [44] confirmed the external-validation gap and the high alert burden. Noseworthy et al. [80] tested an electrocardiogram deep model and found stable performance across race once age and sex were matched. Larrazabal et al. [68] showed that a sex-imbalanced training set lowered diagnostic accuracy for the under-represented sex. Coley et al. [25] reported that a suicide-risk model had lower sensitivity for Black and American Indian patients. Adam et al. [1] found that race-correlated language in clinical notes shifted downstream model recommendations. Pierson et al. [86] trained a pain model on patient-reported scores rather than radiologist labels and explained a larger share of the racial pain gap. Davis et al. [29] tracked an in-use model across eleven years of VA data and observed subgroup performance drift after deployment. Rajkomar et al. [91] and Chen et al. [20] set out fairness desiderata for clinical machine learning. Singh et al. [100] argued that a deployed clinical model needs continual rather than one-off evaluation. Finlayson et al. [38] catalogued the dataset-shift modes that degrade clinical models, including case-mix and practice-pattern change. Vasey et al. [105] and Collins et al. [26] issued the DECIDE-AI and TRIPOD+AI reporting standards, both of which now require subgroup performance reporting. Sculley et al. [96] and Breck et al. [15] documented the monitoring debt that accrues when a model

ships without a test harness. The MIMIC-IV [55] and eICU [89] databases supply the public critical-care streams used for external validation. Texas-100X [53] provides 925 128 hospital discharge records with the demographic mix this paper’s synthesiser reproduces.

2.7 Counterfactual auditing of deployed classifiers

Maughan and Near [74] proposed prediction sensitivity, a per-batch scalar. It estimates how often a counterfactual swap of the protected attribute changes the verdict, without the attribute at inference time. Wachter et al. [109] defined a counterfactual explanation as the smallest input change that flips the decision. Ustun et al. [104] certified whether such a change is attainable through actionable features only. Chen et al. [21] bounded a disparity estimate when the protected attribute is unobserved at audit time. These methods return a scalar or a per-instance explanation at a fixed time. They provide no anytime-valid alarm and no diagnostic decomposition of the flip into a data signal and a model signal. CHRONO-Fair targets that gap.

2.8 Fairness auditing dashboards and post-deployment monitoring tools

A separate line of work has packaged the fairness audit as a tool that an analyst or governance team can run. Aequitas, released by Saleiro and colleagues, exposes group-rate, false-positive-rate, and false-negative-rate parity ratios across protected groups on a fixed prediction file and renders the resulting bias report as a per-attribute web table. Fairlearn, originating from the Microsoft FairLearn project of Bird and colleagues, ships a Python interface that computes selection-rate, equalised-odds, and demographic-parity statistics on a held-out evaluation set together with an in-processing reduction-to-classification mitigator. AIF360, released by Bellamy and colleagues at IBM Research, bundles roughly twenty group and individual fairness metrics with pre-, in-, and post-processing mitigators in one library. The Microsoft Responsible AI Dashboard composes Fairlearn, InterpretML, error-analysis, and counterfactual-explanation components inside a single notebook-embeddable widget aimed at a model developer’s review session. All four artefacts operate in batch on a fixed set of predictions, present a snapshot at the moment of inspection, and offer no anytime-valid statistical guarantee under repeated peeking. The Davis et al. [29] VA fairness-drift study sits on the empirical side of the same gap: it tracks subgroup performance quarterly across eleven years of deployed predictions, demonstrates that the drift is real, and stops short of providing a per-cell sequential alarm at intersectional granularity. The Verdict Flip Rate of [7] is the closest prior artefact to the present work, in that it scores the stability of the fairness verdict itself rather than of a point metric; it does so in batch mode with no time axis. Table 1 summarises this auditing-tool layer along five axes that matter for clinical post-deployment use.

Takeaway 1. The four auditing toolkits in routine use (Aequitas, Fairlearn, AIF360, Microsoft RAI Dashboard) all operate in batch on a fixed prediction file. None of them carries a sequential statistical guarantee under repeated inspection, which is the regime a clinical governance team actually inhabits.

Takeaway 2. The closest streaming clinical evidence, the quarterly VA fairness-drift study of Davis et al., demonstrates that the drift is real but stops at quarterly aggregation and at a single subgroup metric. It leaves open the per-cell, anytime-valid layer that CHRONO-Fair contributes.

Takeaway 3. The Verdict Flip Rate of the prior VFR-Audit work is the closest object to CHRONO-Fair’s monitored quantity, but it is reported as a

Table 1. Fairness auditing dashboards and monitoring tools, against the operating regime a clinical post-deployment monitor needs. Entries cover function, operating mode, output object, and the residual gap that CHRONO-Fair targets. No new statistical claim is made by this table; it consolidates feature descriptions from each tool’s public documentation and from the cited drift- and verdict-stability work.

Tool or work	Main function	Operating mode	Output object	Residual gap for Fair
Aequitas	Group-rate and error-rate parity ratios per protected attribute	Batch on a fixed prediction file	Per-attribute bias report (web table)	No sequential alarm; cell anytime-valid
Fairlearn	Selection-rate, equalised-odds, demographic-parity metrics; reduction-based mitigator	Batch on held-out evaluation set	Metric dashboard plus mitigator object	No streaming verdict; agnostic split
AIF360	Library of 20 group and individual metrics with pre-, in-, post-processing mitigators	Batch on held-out evaluation set	Metric report and corrected model	Inspection-time estimate; no flip-monitoring
Microsoft RAI Dashboard	Fairlearn + InterpretML + counterfactual-explanation widgets in one developer view	Batch in notebook session	Interactive review widget	Aimed at development; not at continuous monitoring
Fairness drift on VA cohort [29]	Quarterly subgroup performance tracking on deployed VA model	Periodic batch (quarterly)	Trend report per subgroup	No anytime-valid alarm; single metric per
Verdict Flip Rate [7]	Static verdict-stability scalar on Texas-100X batch	Static (one batch evaluation)	Single VFR number per protected attribute	No time axis; no sequential alarm

single batch scalar with no time axis. CHRONO-Fair lifts the same quantity into a per-patient hazard curve under an anytime-valid alarm rule.

Takeaway 4. Existing tools report “what is the disparity now”. They do not report “when did it open”, “is the signal larger than noise under repeated peeking”, or “is this a label-pipeline issue or a sample-size issue”. CHRONO-Fair is built around those three questions.

The mapping between each of the four CHRONO-Fair modules and the gap each one closes is direct. The Flip Hazard estimator closes the time-axis gap left open by the four batch toolkits and by static VFR, by reporting a per-arrival-index survival curve rather than a scalar. The per-cell shrunk-GROW e-process with step-wise Benjamini-Hochberg adjustment closes the peeking-penalty gap left open by every batch tool and by the quarterly Davis et al. cadence, by giving a per-inspection-time guarantee under Ville’s inequality. The aleatoric-epistemic decomposition closes the interpretation gap that all of Aequitas, Fairlearn, AIF360, and the Microsoft RAI Dashboard leave open, by attaching a diagnostic triage label (data-side, model-side, or mixed) to every alarmed cell rather than reporting only its magnitude. RCAP closes the regression-output gap that the flip-rate and verdict literature has not addressed, by quantifying counterfactual rank-position shift on a length-of-stay output for which a binary verdict is not defined. This four-way mapping is the contribution map of the paper.

2.9 Positioning against prior work

Table 2 places CHRONO-Fair against eight families of prior work on nine capability axes. The axes are the ones a post-deployment clinical monitor needs. Static or streaming records the

Table 2. Capability comparison of fairness-audit method families. “Y” denotes the capability is present as a designed feature, “–” denotes absent or not reported, and “P” denotes partial support. The final row is the contribution of this paper; the footnote states what each “P” on that row means.

Method family (representative work)	Static/ streaming	Verdict reliability	Anytime-valid per-cell	Cross-cell adjustment	Inter- sectional	CF-flip monitoring	Diagnostic attribution	Regression/ rank-alloc.	Real clinical evidence
Group metrics, batch toolkits [45]	static	–	–	–	P	–	–	–	various
Bootstrap-CI fairness audit [22]	static	Y	–	–	–	–	–	–	–
Predictive multiplicity, eDFR [73, 77]	static	Y	–	–	–	–	–	–	some
Verdict Flip Rate [7]	static	Y	–	–	P	Y	–	–	Texas-100X batch
Prediction sensitivity [74]	per batch	–	–	–	–	Y	–	–	–
Online fair learning [50]	streaming	–	–	–	–	–	–	–	–
Fairness-drift monitoring [29]	quarterly	P	–	–	P	–	–	–	VA cohort
Anytime-valid risk tracking [88]	streaming	–	Y	–	–	–	–	–	–
CHRONO-Fair (this paper)	streaming	Y	Y	P	Y	Y	P	Y	P

Footnote. For the CHRONO-Fair row, “P” on cross-cell adjustment means an inspection-time step-wise Benjamini-Hochberg adjustment rather than a full time-uniform online false-discovery procedure. “P” on diagnostic attribution means an aleatoric/epistemic triage signal, not a confirmed causal root cause. “P” on real clinical evidence means a real Texas-100X batch verdict-stability analysis combined with Texas-100X-calibrated controlled streaming and replay experiments, not prospective deployment evidence.

operating mode. Verdict reliability records whether the method reports the stability of a pass or fail verdict. Anytime-valid per-cell monitoring records a false-alarm rate bounded under inspection at every step. Cross-cell adjustment records a multiple-testing correction across monitored cells. Intersectional records cells defined by two or more protected attributes. Counterfactual-flip monitoring records tracking of verdict flips under a protected-attribute counterfactual. Diagnostic triage output records whether the framework attributes a detected gap to a likely data, model, or mixed signal source. Regression support records handling of a continuous, rank-allocation output. Real clinical evidence records the empirical clinical data used.

Read row by row, the table shows that the gap is structural rather than incidental, since no prior method family holds more than two of the nine capabilities at once. The nearest entries each fall short on a different axis. Anytime-valid risk tracking [88] is streaming and anytime-valid, but it monitors aggregate loss rather than a group-conditional fairness verdict and offers no diagnostic attribution. Fairness-drift monitoring [29] is the closest clinical work, yet it operates on a quarterly cadence, monitors a single metric, and gives no statistical guarantee under repeated inspection. The Verdict Flip Rate [7] measures verdict instability directly but collapses it to a scalar with no time axis. CHRONO-Fair occupies the remaining position by combining the streaming, anytime-valid, intersectional, and diagnostic capabilities, while the three partial marks in its row record exactly where its support is qualified rather than complete.

2.10 Open gaps addressed by this paper

Four gaps follow from Table 2. *Gap 1, the time axis.* A batch verdict cannot say when a fairness gap opened during deployment, and a clinical response needs that timestamp. *Gap 2, the peeking penalty.* A governance team inspects a monitor often. A fixed-sample test inflates its false-alarm rate under repeated looks. A continuous monitor therefore needs an anytime-valid guarantee. *Gap 3, the diagnostic interpretation.* A detected gap with no diagnostic interpretation does not tell a team whether to fix the data pipeline or to collect more samples. *Gap 4, regression outputs.* Length-of-stay and cost models produce continuous outputs. The flip-rate and verdict literature is built for binary classification. CHRONO-Fair addresses the four gaps with the Flip Hazard estimator, the e-process, the aleatoric-epistemic decomposition, and RCAP, in that order.

3 PROBLEM FORMULATION

3.1 Clinical fairness, the fairness verdict, and the flip indicator

This paper uses three terms with fixed meaning throughout. *Clinical fairness* denotes the property that the thresholded decisions of a deployed clinical machine learning model do not depend on a patient’s legally or ethically protected attributes beyond what is clinically justified, evaluated by an agreed group-fairness metric and threshold (for example disparate impact above 0.8). The *fairness verdict* is the binary pass-or-fail outcome of that evaluation at a given threshold. The *counterfactual verdict flip*, or *verdict flip* for short, is the event in which the verdict changes between the model’s factual decision on a patient and its counterfactual decision under a swap of the protected attribute. The framework treats the verdict flip as the observable governance object and monitors its rate as patients arrive.

3.2 Notation and assumptions

Let $\{(X_i, A_i, Y_i)\}_{i \geq 1}$ be a stream of patient records, where $X_i \in \mathbb{R}^d$ are non-sensitive features, A_i is a protected attribute taking values in a finite set $\mathcal{A}$, and $Y_i \in \{0, 1\}$ or $Y_i \in \mathbb{R}_{\geq 0}$ is the label. The arrival index t orders patients by admission time. The deployed model at index t is $f_t: (X, A) \rightarrow \hat{Y}$. The model may change between two indices through periodic retraining, threshold recalibration, or hot-fixes. Re-training events are exposed to CHRONO-Fair as version stamps. Between two retrains, the model is held fixed and the framework monitors the decision stream rather than the model parameters.

Let A_i^{cf} denote a counterfactual sensitive-attribute value drawn from $\mathcal{A}$. In this paper, A_i^{cf} is fixed to a reference level (typically the majority group). The counterfactual prediction $\hat{Y}_i^{\text{cf}} = f_t(X_i, A_i^{\text{cf}})$ is obtained by setting the protected attribute to its counterfactual value and removing the marginal effect on observable features through a feature-shift adjustment. This corresponds to the deployment-time counterfactual of Maughan and Near [74]. It is weaker than the SCM-based counterfactual of Kusner et al. [65] but it is computable at inference time without an audited causal graph.

3.3 Flip indicator

For binary classification with threshold τ , the flip indicator is

$$Z_i = \mathbf{1}\left\{\mathbf{1}\left\{\hat{Y}_i \geq \tau\right\} \neq \mathbf{1}\left\{\hat{Y}_i^{\text{cf}} \geq \tau\right\}\right\}. \quad (1)$$

For regression, a threshold-free analogue is the counterfactual rank shift formalised in Section 4.4. The flip indicator depends on f_t at the patient’s arrival index, not on f_t at the reporting time, so retraining events do not retroactively invalidate prior Z_i .

3.4 Baseline calibration

Each cell c defined by a value of the protected attribute has a baseline flip rate $\rho_0^{(c)}$. The baseline is calibrated on the pre-deployment validation set used to lock the model for release. The validation set should match the deployment population and contain at least 500 patients per cell. If the per-cell calibration size falls below 500, $\rho_0^{(c)}$ is set to the marginal flip rate on the same validation set and a wider warm-up window is used. Calibration is repeated each time the model is replaced.

4 METHOD

CHRONO-Fair is one framework with four coordinated estimators that share a single counterfactual flip indicator. Figure 2 shows the data flow. A patient stream feeds the

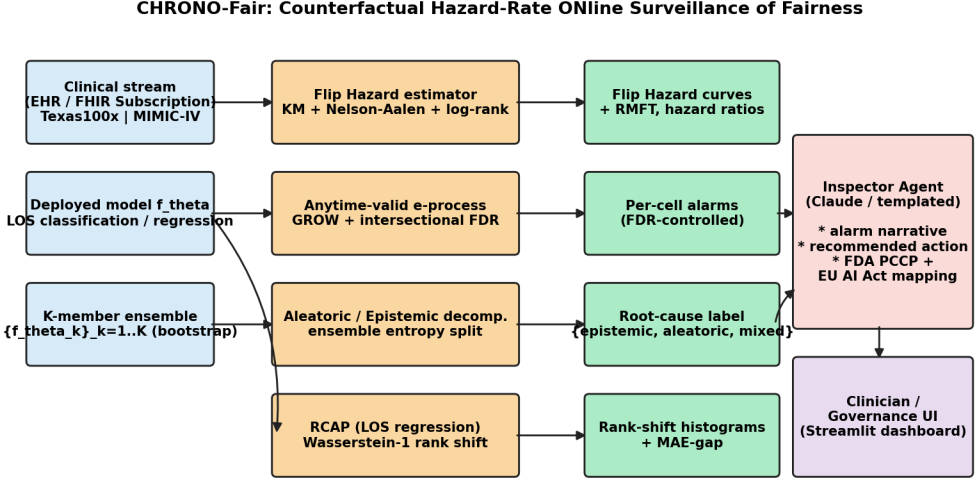


Fig. 2. CHRONO-Fair architecture. A clinical stream feeds a deployed model and a bootstrap ensemble. Four estimators run concurrently: Flip Hazard survival, anytime-valid e-process, aleatoric/epistemic decomposition, and RCAP. The Inspector Agent translates alarms into a structured governance report.

deployed model and its counterfactual, the flip indicator is computed, and the four estimators consume it. The Flip Hazard estimator and the e-process are sequential. The decomposition and RCAP run on demand. The Inspector report aggregates the four outputs for governance review.

4.1 Flip Hazard estimator

Definition 4.1 (Flip Hazard). For protected group a , the Flip Hazard function is

$$\lambda_a(t) = \lim_{\Delta \rightarrow 0} \frac{1}{\Delta} \Pr(Z_t = 1, T \geq t \mid A = a) / \Pr(T \geq t \mid A = a), \quad (2)$$

where $T = \inf\{t : Z_t = 1\}$ is the first flip time and the limit is taken on the patient-arrival index.

The no-flip survival is $S_a(t) = \exp(-\int_0^t \lambda_a(s) ds)$. The scalar VFR of [7] corresponds to the aggregate $1 - S_a(\tau^*)$ at the full audit horizon τ^* . The framework reports survival as a curve, not a scalar.

The estimator is non-parametric, so it makes no distributional assumption on the flip-time. Kaplan-Meier survival [58] and Nelson-Aalen cumulative hazard [79] are computed per group, and Greenwood confidence bands are produced around each survival curve. The pairwise log-rank test [5] is applied for between-group comparisons. The Restricted Mean Flip Time at horizon τ^* is

$$\text{RMFT}_a(\tau^*) = \int_0^{\tau^*} S_a(t) dt, \quad (3)$$

and admits a closed-form step-integration on the empirical KM curve.

4.2 Anytime-valid e-process

The e-process for one cell c tests $H_0: \mathbb{E}[Z_t] = \rho_0^{(c)}$ against the one-sided alternative $\mathbb{E}[Z_t] > \rho_0^{(c)}$:

$$E_t^{(c)} = \prod_{s=1}^t (1 + \lambda_s^{(c)}(Z_s - \rho_0^{(c)})), \quad (4)$$

where $\lambda_s^{(c)}$ is a predictable bet sequence measurable in $\sigma(Z_1, \dots, Z_{s-1})$. The bet is clipped to $[-1/(1 - \rho_0), 1/\rho_0]$ to guarantee $E_t^{(c)} \geq 0$. The framework uses a shrunk GROW rule:

$$\lambda_s = c \frac{\bar{Z}_{s-1} - \rho_0}{\rho_0(1 - \rho_0)} \cdot \mathbf{1}\{s \geq n_{\text{warm}}\}, \quad (5)$$

with shrinkage $c = 0.25$ and warm-up $n_{\text{warm}} = 100$. The shrinkage and warm-up are conservative choices that trade detection speed for empirical false-alarm control; they are not load-bearing for anytime validity, only for finite-sample calibration on small streams.

THEOREM 4.2 (ANYTIME-VALID TYPE-I CONTROL). *Under H_0 , the process $(E_t^{(c)})_{t \geq 0}$ is a non-negative supermartingale with $\mathbb{E}[E_t^{(c)}] \leq 1$. By Ville’s inequality, for any stopping time τ ,*

$$\Pr(\sup_{t \leq \tau} E_t^{(c)} \geq 1/\alpha \mid H_0) \leq \alpha. \quad (6)$$

The proof follows the supermartingale construction in Ramdas et al. [92], applied to the bounded random variable $Z_s - \rho_0 \in [-\rho_0, 1 - \rho_0]$.

Intersectional multiple-testing control. For m disjoint cells, the framework runs m parallel monitors. At each inspection step, step-wise Benjamini-Hochberg [10] is applied to the e-value-implied p -values $p_t^{(c)} = 1/E_t^{(c)}$ clipped to $[0, 1]$.

The current multiple-testing layer should be interpreted as cross-sectional adjustment over the monitored cells at each inspection time, not as a full time-uniform online false-discovery guarantee. The per-cell anytime-valid Type-I control of Theorem 4.2 holds for each cell separately. The cross-sectional step-wise Benjamini-Hochberg adjustment controls the false-discovery proportion among the cells inspected at a given step, under exchangeability of cell e-values. Extending CHRONO-Fair with a time-uniform online false-discovery procedure, such as LORD [52], SAFFRON, alpha-investing [39], or e-BH, is future work. The framework therefore reports findings as “flagged at inspection step t ” rather than “time-uniformly significant”.

Complexity. Per arriving patient, the e-process update is $O(1)$ per cell. The step-wise BH sort across m cells is $O(m \log m)$. For a four-attribute deployment, m ranges from 13 marginal cells to 80 full intersectional cells. End-to-end latency on a single CPU core is below 5 ms per patient.

4.3 Aleatoric and epistemic decomposition

Given a bootstrap ensemble of K classifiers with member probabilities $\{p_k(x)\}_{k=1}^K$ and counterfactual member probabilities $\{p_k^{\text{cf}}(x)\}_{k=1}^K$, the per-instance flip mass is

$$\bar{p}^{\text{flip}}(x) = \frac{1}{K} \sum_{k=1}^K \mathbf{1}\{(p_k(x) \geq \tau) \neq (p_k^{\text{cf}}(x) \geq \tau)\}. \quad (7)$$

The total predictive entropy admits the decomposition of Depeweg et al. [30] into

$$H[\bar{p}] = \underbrace{\mathbb{E}_k H[p_k]}_{\text{aleatoric}} + \underbrace{(H[\bar{p}] - \mathbb{E}_k H[p_k])}_{\text{epistemic}}. \quad (8)$$

For each instance, the flip mass is apportioned in proportion to the aleatoric and epistemic shares of the total entropy. Aggregated over a cell, the report summarises whether the flip signal is more consistent with data-driven noise (ensemble members that are individually confident and agree), with model uncertainty (members that disagree on the verdict), or with a mixed pattern (intermediate shares). The decomposition is intended as a diagnostic triage signal for governance review rather than as a causal attribution. Section 6.3 validates the formula on controlled inputs (Track 1) and reports the mixed-pattern behaviour of realistic ensembles (Track 2); a mixed triage label is the appropriate output for a real pipeline in which both data and model effects coexist, rather than evidence of failure.

4.4 Regression Counterfactual Allocation Parity

Let $f(x, a)$ be a regressor producing a continuous output, length of stay in days in this paper. Let $\hat{F}_{\hat{Y}|A=a}$ be the empirical cumulative distribution function of the predicted output within group a . For patient i in group a , the predicted-rank position under the factual and the counterfactual attribute is

$$u_i(a) = \hat{F}_{\hat{Y}|A=a}(f(x_i, a)), \quad (9)$$

$$u_i(a') = \hat{F}_{\hat{Y}|A=a'}(f(x_i, a')). \quad (10)$$

Each u_i lies in $[0, 1]$ and gives the patient’s percentile position in the group-conditional predicted-output distribution. The per-patient rank shift is $\Delta_{\text{rank}}(x_i, a, a') = u_i(a) - u_i(a')$. The aggregate RCAP statistic is the Wasserstein-1 distance between the two rank-position distributions

$$\text{RCAP}(a, a') = W_1(\{u_i(a) : i \in a\}, \{u_i(a') : i \in a'\}). \quad (11)$$

A bootstrap 95% confidence interval is computed by resampling the per-patient pairs $(u_i(a), u_i(a'))$ $B = 1000$ times.

RCAP measures how far a patient’s predicted length-of-stay rank position shifts under the protected-attribute counterfactual. The rank position, not the raw error alone, determines prioritisation in length-of-stay-based bed allocation. The group-MAE gap collapses a directional, distribution-aware allocation shift into the average regression error and does not register an allocation-only change. RCAP is reported on the $[0, 1]$ rank scale, so a value of 0.0163 corresponds to a shift of 1.63 percentile points.

4.5 Inspector report

When the intersectional monitor flags a cell, the framework emits a structured report. Fields include: cell label, observed n , current flip rate, baseline flip rate, e-value, hazard ratio against baseline, aleatoric and epistemic shares, dominant component, recommended action, and a suggested regulatory mapping. The recommended action is derived deterministically from the dominant component: an aleatoric-dominated cell triggers a label-pipeline audit recommendation; an epistemic-dominated cell triggers a recommendation to enlarge the stratum sample. The regulatory mapping is a non-binding cross-reference to FDA PCCP [103] sections, EU AI Act [37] articles, and STANDING Together [3] recommendations, intended to accelerate review by clinical governance staff. Formal regulatory validation is future work.

4.6 Relationship to VFR-Audit and monitored model setup

CHRONO-Fair is not a fairness mitigator. It is a post-deployment monitoring framework. It can monitor any deployed clinical machine learning model after a pre-release fairness audit. The deployed model may be released without mitigation, or after reweighing, threshold adjustment, in-processing constraints, or any other pre-release method. In this paper, VFR-Audit [7] is cited only as the prior static audit framework and conceptual predecessor. VFR-Audit measures whether thresholded fairness verdicts are reliable under three sources of variation. The three sources are resampling, audit-size change, and cross-hospital site change. VFR-Audit introduces the Verdict Flip Rate as a static scalar summary of that reliability. CHRONO-Fair extends that idea by monitoring whether counterfactual fairness-verdict flips emerge over time after deployment.

The streaming experiments in Section 6 use a pre-release fairness-adjusted model generated for the present experiments. The pre-release adjustment is part of the experimental configuration of this paper. It is not a named contribution and not a method introduced by the VFR-Audit paper. The monitored-model setup is reported in Section 6.10, where the pre-release fairness-adjustment numbers are tabulated.

CHRONO-Fair does not require any specific pre-release mitigation method. It monitors the stream of factual and counterfactual predictions produced by the deployed model. The same monitor therefore applies after reweighing, threshold adjustment, in-processing mitigation, or no mitigation at all. The baseline flip rate ρ_0 in Section 3 absorbs whichever pre-release configuration the deployment site used. The framework is in that sense agnostic to the upstream pipeline.

5 DEPLOYMENT

The reference implementation is open-sourced as a Python 3.11 package with no GPU dependency.

5.1 Software stack

Core libraries are NumPy, pandas, scikit-learn, SciPy, matplotlib, and statsmodels. The dashboard layer uses Streamlit. No external API calls are made on the critical path.

5.2 Data interface

The framework consumes a stream of records carrying an arrival timestamp, a patient identifier, the factual and counterfactual predicted labels, the optional predicted regression value, the protected attributes, and an optional model version stamp. A reference adapter for standard clinical interoperability formats and for batch JSONL is provided with the released code. Specific transport-layer choices are deployment-site decisions and are not part of the framework.

5.3 Latency and failure modes

The per-patient update path is well below 10ms on a single CPU core in the released implementation, so the monitor is unlikely to be the bottleneck in any realistic admission stream. The framework handles missing protected attributes by dropping the affected record from the intersectional monitor while the marginal monitor continues, an ensemble crash by skipping the decomposition while the e-process continues, and a corrupted counterfactual by recording the indicator as missing and extending the warm-up window.

5.4 PHI handling and configuration

The framework receives de-identified records by contract and does not store raw protected health information. Audit logs are append-only and contain only e-values, cell labels, and timestamps. Default operating points are $\alpha = 0.05$ for the anytime-valid threshold, $q = 0.10$ for the inspection-time Benjamini-Hochberg level, $n_{\text{warm}} = 100$ for the e-process warm-up, $c = 0.25$ for the GROW shrinkage, $K = 11$ ensemble members, and $\tau^* = 5000$ patients for the RMFT horizon. Further implementation details, including the cadence schedule, the messaging-channel adapter, and the seven-field record schema, are documented in the released code and the supplementary material rather than in the main text.

5.5 Scope statement

The deployment description above specifies the contract of a research prototype, not the certification of a production system. Hospital integration would require institutional review-board approval, local validation on the deployment site’s population, and human-in-the-loop review of every alarm. The framework is not a medical device.

6 EXPERIMENTS

The evidence base of this study is organised as three layers that decline in directness from real clinical data toward controlled simulation. Layer 1 is the real Texas-100X batch verdict-stability analysis of Section 6.8. Layer 2 is the set of Texas-100X-calibrated controlled streaming and retrospective replay experiments that characterise sequential monitoring behaviour under known drift settings, including the quarter-based and hospital-group retrospective replays of Section 6.16, which approximate deployment ordering on the real cohort but do not replace live clinical validation. Layer 3 is a robustness and stress-test layer covering baseline miscalibration, small intersectional cells, and label delay. Only Layer 1 uses real clinical records directly; Layers 2 and 3 evaluate sequential behaviour under controlled replay or simulation. Texas-100X is administrative discharge data rather than a live admission stream, so prospective EHR-stream validation remains future work.

What the framework is compared against, and where each comparator comes from. The post-deployment monitoring comparison is run on a single shared substrate: the same Texas-100X-calibrated stream, the same pre-corrected deployed model (Section 4.6), the same patient-level counterfactual flip indicator. Holding the substrate fixed means the reported gap between methods reflects the alarm rule, not the data path. Five comparator rules are drawn from named published instruments, and each enters the comparison as published rather than re-derived.

The snapshot reference is the scalar Verdict Flip Rate of the prior work [7], computed once on a pre-deployment validation window. It is the simplest possible monitor: a single number, never refreshed after release. Its purpose is to show how much information a deployment loses by not updating at all. Sitting one step above the snapshot is the quarterly-audit instrument of Davis et al. [29], here reproduced as a two-proportion z -test on non-overlapping batches of 500 patients. The Davis paper ran its test at a calendar-quarter cadence on a VA cohort; the same statistical object is used here, with the cadence converted to a patient count to fit a single-site deployment. The change-point family is represented by ADWIN of Bifet and Gavalda [12], run on the per-patient flip indicator at two inspection schedules to separate cadence effects from rule effects. A fairness-gap variant of ADWIN, in the spirit of the JAMIA 2025 fairness-drift recommendation [29], applies the same detector to the centred flip series rather than to the raw indicator. The closest direct precedent on the counterfactual

Experiment	n	Trials	Drift setting	Metric	Baseline(s)
Exp. 1 detection delay	10 000	50	onset at $t = 5\,000$; $\rho_0 = 0.05 \rightarrow \rho_1 = 0.15$	patients-to-alarm; detection rate	ADWIN ($k=1$, $k=25$), Davis-ADWIN, periodic batch (500), static VFR
Exp. 2 decomposition	1 000	30/scenario	3 scenarios (aleatoric, epistemic, mixed)	confusion-matrix accuracy	none (validation against ground truth)
Exp. 3 Flip Hazard	15 000	1	no drift, structural disparity	RMFT, log-rank χ^2 , KM survival	scalar VFR [7]
Exp. 4 RCAP	20 000	1 (1 000 bootstraps)	no drift, structural disparity	Wasserstein-1 rank shift, MAE gap	group-MAE gap
Exp. 5 false-alarm	10 000	200/ α	no drift (H_0)	empirical false-alarm rate per nominal α	nominal α line
Exp. 6 sensitivity	12 000	30/ ρ_1	onset at $t = 6\,000$; $\rho_1 \in [0.06, 0.20]$	detection rate; mean delay vs drift size	none (characterisation)
Exp. 7 four-attribute audit	15 000	1	no drift; structural racial disparity	log-rank χ^2 across race, ethnicity, sex, age	marginal flip rate
Exp. 8 real Texas-100X VFR	925 128	1000 boot.	real audit (no simulation)	point vs CI vs VFR verdict stability	point estimate; bootstrap CI
Exp. 9 robustness stress tests	200 to 12 000	100/setting	misalignment, small cell, label delay	false-alarm rate; detection rate; delay	correctly-calibrated monitor
Exp. 10 multi-model fairness	925 128	1	real test set; LR, RF, GB, XGBoost, LightGBM, DNN	DI, WTPR, SPD, EOD, PPV across 4 attrs	six models
Exp. 11 temporal/hospital replay	30 000; 67 330	1	quarter drift; hospital case-mix; real record order	alarm; detection delay	ADWIN; static VFR
Exp. 12 pre-release fairness-adjustment setup	925 128	1	monitored-model configuration for this paper	DI / WTPR per attribute on the pre-release adjusted model	uncorrected base classifier

Table 3. Experiment summary. Each row lists stream length, replication count, drift configuration, primary metric, and baselines.

side is Prediction Sensitivity [74], which reports a per-batch counterfactual-flip rate without an alarm rule; it appears in the comparison as a methods-level reference, since the published version is an audit rather than a sequential alarm.

The framework introduced in this paper sits beside those rules rather than replacing them. Its statistical content is the per-cell shrunk-GROW e-process under Ville’s inequality with a step-wise Benjamini-Hochberg cross-cell layer at each inspection time, plus the diagnostic aleatoric-epistemic split and the RCAP rank-allocation statistic. Holding the deployed model and the flip indicator constant across all six rules, the post-deployment comparison in Section 6.2 therefore reads as a head-to-head of alarm rules under matched conditions.

The monitored model used in the streaming experiments is described in Section 6, where the pre-release fairness-adjustment setup is reported as part of the experimental configuration. The pre-release adjustment is not a published method from any prior work; it is a configuration of the present experiments that produces the deployed-model stream that CHRONO-Fair consumes. A cross-study numerical comparison against other published length-of-stay fairness work was not undertaken because no prior length-of-stay study runs a per-cell anytime-valid monitor or a Flip Hazard estimator on the same substrate, which would make a head-to-head number meaningless.

Table 3 summarises each experiment.

6.1 Synthesiser

A synthetic stream calibrated to the Texas-100X record statistics [53] was generated. The mix matched the dataset’s published demographics: White 48%, Hispanic 30%, Black 13%, Asian/PI 5%, Other 4%; Female 55%, Male 45%; Non-Hispanic 65%, Hispanic 35%; age groups Pediatric 5%, Young Adult 20%, Middle-aged 40%, Elderly 35%. Continuous LOS was generated as $\text{LOS} = 4.2 + \beta^\top X + \epsilon$ with $\epsilon \sim \mathcal{N}(0, 0.5^2)$. The binary target $y_{\text{ext}} = \mathbf{1}\{\text{LOS} > 3\}$ matched the prior implementation’s class balance of 55% normal stay, 45% extended stay.

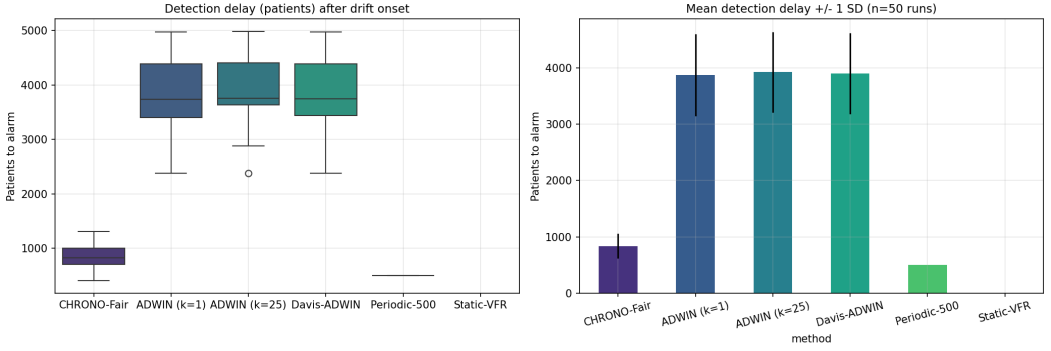


Fig. 3. Section 6.2. Detection delay across 50 Monte-Carlo runs at $n = 10\,000$, drift onset at $t = 5\,000$. CHRONO-Fair: 50 of 50 alarms at mean delay 837. ADWIN variants and Davis-ADWIN: 21 of 50 at mean delay 3878 to 3927. Periodic batch (500): 24 of 50 at mean 499 without peeking control. Static VFR: 0 of 50.

A structural disparity of $+0.4$ on the first three risk features was injected for Black and Hispanic groups, reproducing the bias structure documented in the prior VFR-Audit work. Optional knobs controlled drift onset and aleatoric label noise.

6.2 Detection delay

Setup. A stream of $n = 10\,000$ patients was generated under $\rho_0 = 0.05$ until index $t = 5\,000$ and under $\rho_1 = 0.15$ thereafter. The CHRONO-Fair e-process was run alongside four baselines: ADWIN with check-every-patient ($k = 1$), ADWIN with check-every-25-patients ($k = 25$), a Davis-style ADWIN on the binary fairness gap series [29], a periodic two-proportion z -test on non-overlapping batches of 500, and a static VFR snapshot computed once before deployment. Fifty Monte-Carlo replications were run with seeds 0 to 49.

Metric. For each method and replication, the number of patients between drift onset at $t = 5\,000$ and the first alarm was recorded. Runs that produced an alarm before drift onset were counted as false alarms and excluded from the delay calculation.

Result. CHRONO-Fair alarmed in 50 of 50 replications with a mean delay of 837 patients (median 828, standard deviation 221). ADWIN at $k = 1$ alarmed in 21 of 50 replications at mean delay 3878 patients. ADWIN at $k = 25$ alarmed in 21 of 50 at mean 3927. Davis-ADWIN alarmed in 21 of 50 at mean 3904. The periodic batch test alarmed in 24 of 50 at mean 499 patients but did not control false alarms across repeated peeks. Static VFR did not alarm. Under a separate no-disparity sanity scenario at $n = 10\,000$ and seeds 10000 to 10049, CHRONO-Fair raised 0 alarms in 50 replications.

Interpretation. The detection-rate gap (50 of 50 against 21 of 50) is larger than the delay gap and is the headline result. ADWIN’s coverage at $k = 1$ and $k = 25$ is similar, indicating that the framework’s advantage is not an artefact of check-frequency. The periodic batch test trades coverage for delay and does not satisfy the FDA PCCP expectation of continuous monitoring [103].

Figure 3 plots the per-method detection delay across the 50 runs.

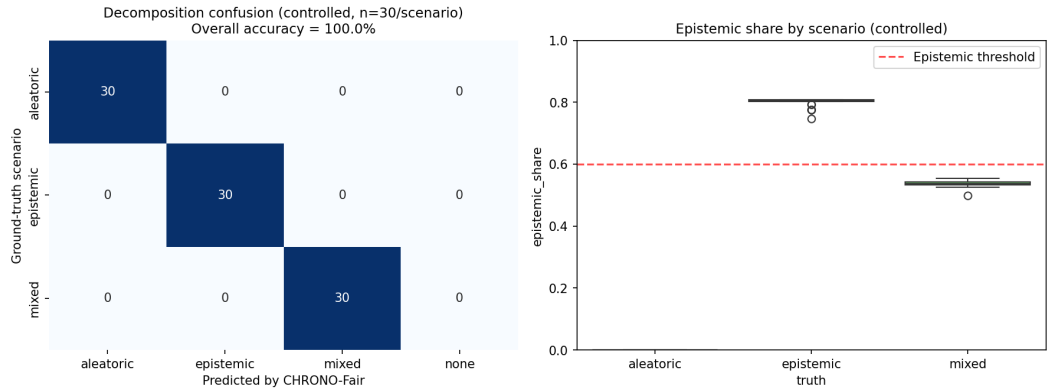


Fig. 4. Section 6.3 Track 1. Controlled decomposition validation, 30 trials per scenario at $n = 1000$. Confusion matrix on the left; per-scenario epistemic share on the right. The 0.6 threshold is the Inspector Agent decision boundary.

6.3 Decomposition accuracy

Setup. Two tracks were run. Track 1 generated ensembles of 15 binary classifiers whose per-instance probabilities were constructed by design to be aleatoric-dominated, epistemic-dominated, or mixed. Thirty replications per scenario were generated with seeds 0 to 29. Track 2 trained ensembles of 11 single decision trees with depth 6, each on a disjoint slice of the minority training data (epistemic by construction) or the full training data with 0.25 aleatoric label flip (aleatoric by construction). Fifteen replications per scenario were run.

Metric. The Inspector Agent label (one of **aleatoric**, **epistemic**, **mixed**, **none**) was compared against the ground-truth scenario. The aggregate confusion matrix and accuracy were reported.

Result. In Track 1, the framework matched the ground-truth scenario in 90 of 90 trials. In Track 2, the ML-driven setup, the framework labelled the dominant diagnostic signal as **aleatoric** in 3 of 15 aleatoric trials, 1 of 15 epistemic trials, and 1 of 15 mixed trials; the remaining trials were labelled **mixed**. Net Track 2 accuracy was 37.8%.

Interpretation. The decomposition formula is correct on inputs that satisfy the controlled-ensemble assumption (Track 1). In realistic ensembles trained on overlapping data and audited at deployment time, the dominant component is frequently **mixed**. The framework’s recommended action under **mixed** is parallel data-pipeline audit and ensemble-stabilised retraining, which is the same action a clinical safety officer would take when the diagnostic interpretation is unclear. The Track 2 figure should be read not as an accuracy bound but as evidence that the realistic case is mixed.

Figure 4 plots the controlled-validation confusion matrix and the per-scenario epistemic share.

6.4 Flip Hazard survival on Texas-100X-style data

Setup. A stream of $n = 15000$ patients was generated with seed 42. The deployed classifier was a logistic regression trained on five risk features. Counterfactual predictions were generated by removing the +0.4 minority shift on features 0 to 2. The Flip Hazard estimator was run per protected group with a horizon $\tau^* = 5000$.

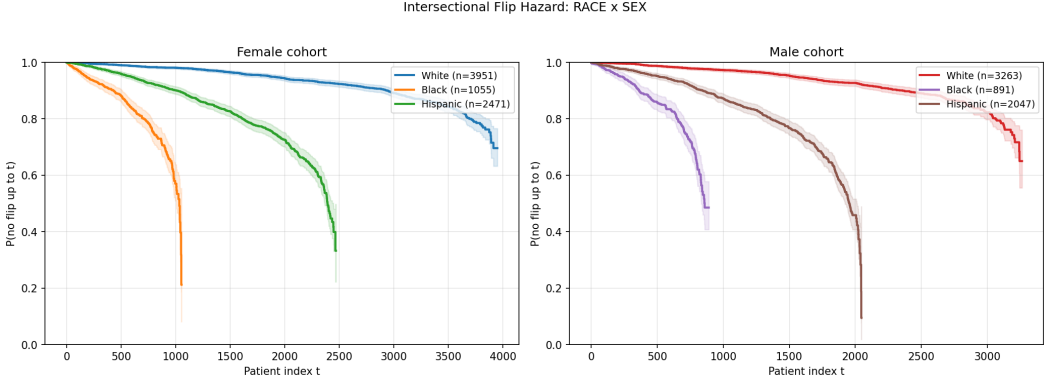


Fig. 5. Section 6.4. Intersectional Flip Hazard: race by sex.

Metric. Per-group Kaplan-Meier no-flip survival curves with Greenwood bands, RMFT at $\tau^* = 5000$, and pairwise log-rank statistics against the White reference group were reported.

Result. RMFT was 4817 patients for White ($n = 7214$), 3940 for Other ($n = 588$), 3837 for Hispanic ($n = 4518$), 3337 for Asian/PI ($n = 734$), and 2459 for Black ($n = 1946$). The scalar VFR was 0.078 for White, 0.066 for Other, 0.201 for Hispanic, 0.065 for Asian/PI, 0.190 for Black. Log-rank statistics against White were $\chi^2 = 1675.49$ (Black), $\chi^2 = 1248.21$ (Hispanic), $\chi^2 = 456.33$ (Asian/PI), $\chi^2 = 455.75$ (Other), all with $p < 10^{-10}$.

Interpretation. The scalar VFR collapses the Black-Hispanic ordering (0.190 against 0.201). The RMFT preserves the ordering: Black has the lowest RMFT (2459), then Asian/PI (3337), then Hispanic (3837), then Other (3940), then White (4817). RMFT distinguishes early-clustering disparity (Black: a flip cluster appears within the first 2500 arrivals) from late-clustering disparity (Hispanic: flips spread over a longer arrival window). The clinical meaning is the same in both cases (disparate exposure), but the operational response differs: an early cluster triggers an immediate review, a late cluster triggers a slower investigation.

Figures 6 to 7 plot the group-stratified survival curves, the intersectional curves, and the RMFT comparison against scalar VFR.

6.5 RCAP for length-of-stay regression

Setup. A stream of $n = 20000$ patients was generated with seed 11. A Ridge regressor (regularisation $\alpha = 1.0$) was trained on the first 8000 records to predict LOS. The remaining 12000 were used for audit. RCAP was computed pairwise across the five racial groups with $B = 1000$ bootstrap replicates for the confidence interval. The group-MAE gap against the White reference was computed as a baseline.

Metric. Wasserstein-1 distance between counterfactual rank distributions per group pair, MAE gap against White, and per-patient rank shift histograms.

Result. RCAP-against-White was computed as the Wasserstein-1 distance between the factual and counterfactual rank-position distributions, with 1000 bootstrap resamples per group. The empirical Wasserstein-1 plug-in is biased upward at small sample size, so the plug-in estimate, the bootstrap mean, and the percentile interval are reported separately rather than as a symmetric interval around the plug-in. Table 4 gives the four values per group. For Black ($n = 1488$) the plug-in W_1 was 0.0163, the bootstrap mean was 0.0197, and the 95% percentile interval was [0.0088, 0.0336]. For Other ($n = 499$) the plug-in was

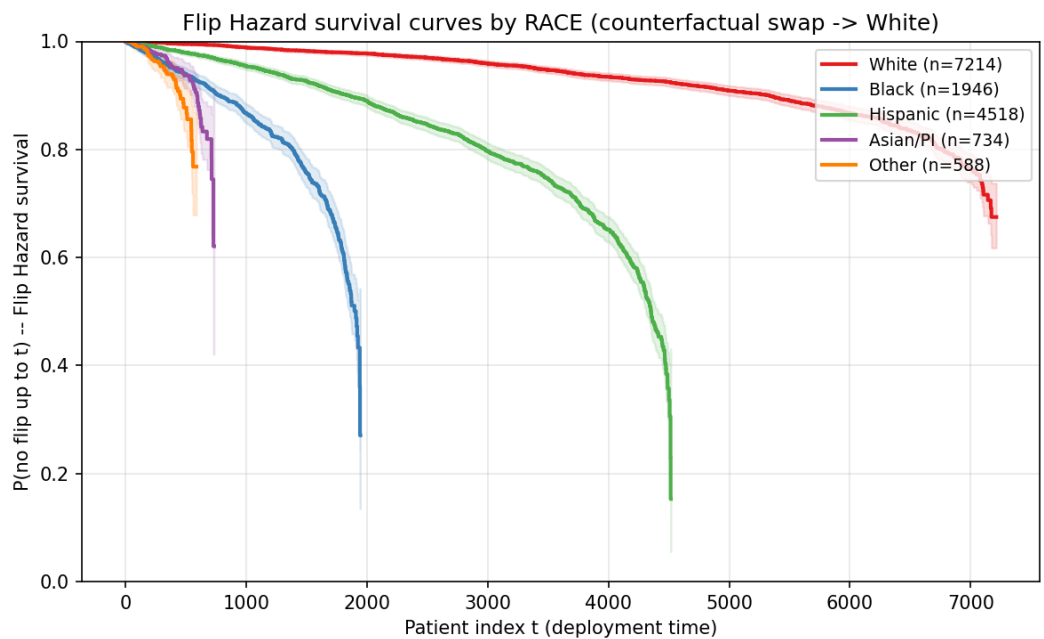


Fig. 6. Section 6.4. Group-stratified Flip Hazard no-flip survival at $n = 15\,000$. Greenwood 95% bands shown.

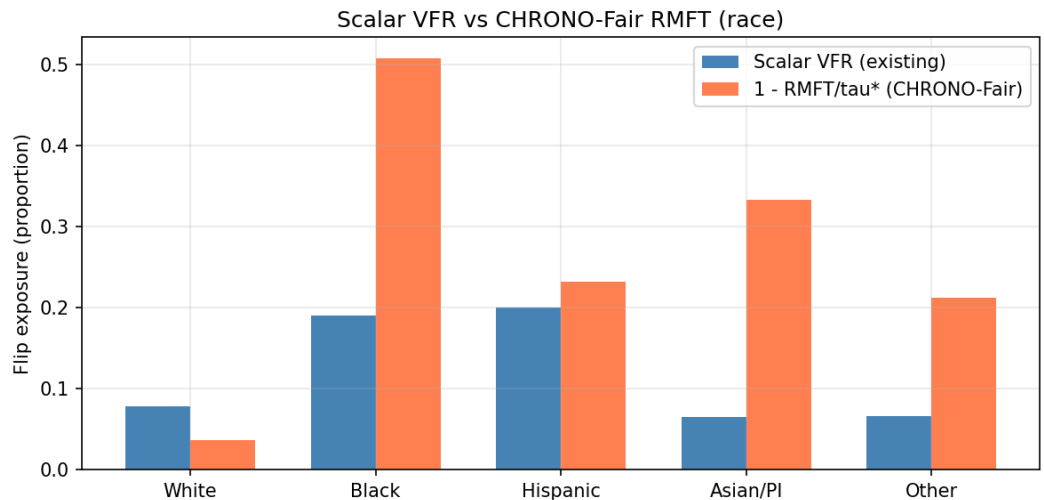


Fig. 7. Section 6.4. Scalar VFR (left bars) against $1 - \text{RMFT}/\tau^*$ (right bars). RMFT preserves the Black versus Hispanic ordering that the scalar VFR collapses.

0.0049, the bootstrap mean was 0.0208, and the interval was $[0.0100, 0.0429]$; the percentile interval lies above the plug-in because the upward bias of the empirical estimator dominates at $n = 499$. The MAE gap against White was 0.0065 days for Black, 0.0062 for Asian/PI,

Table 4. RCAP against the White reference. The plug-in is the Wasserstein-1 distance on the observed rank positions; the bootstrap mean and the 95% percentile interval come from 1 000 resamples. The bootstrap mean exceeds the plug-in for every group, which quantifies the upward small-sample bias of the empirical Wasserstein-1 estimator.

Group	n	Plug-in W_1	Boot. mean	95% interval
Black	1 488	0.0163	0.0197	[0.0088, 0.0336]
Asian/PI	552	0.0142	0.0233	[0.0117, 0.0425]
Hispanic	3 643	0.0066	0.0099	[0.0052, 0.0169]
Other	499	0.0049	0.0208	[0.0100, 0.0429]

0.0006 for Hispanic, and 0.0028 for Other. The MAE ordering was Black, Asian/PI, Other, Hispanic, while the plug-in RCAP ordering was Black, Asian/PI, Hispanic, Other; the two orderings agreed on the leader, Black, but disagreed on the placement of Hispanic and Other.

Interpretation. The MAE gap measures average error, which a regression model can keep low even while systematically shifting one group’s allocation rank. RCAP captures the rank shift directly. The clinical reading is that, under the counterfactual swap, the Black rank-position distribution shifts by 1.63 percentile points from the White-referenced distribution on the plug-in estimate, with a 95% percentile interval of 0.88 to 3.36 percentile points that excludes zero. The triage and bed-allocation literature treats predicted LOS as an ordinal allocation signal, so a rank-position shift is the policy-relevant quantity [101]. Because the empirical Wasserstein-1 estimator is upward-biased at small n , RCAP must be read together with the per-cell sample size, and cells near $n = 500$ such as Other should be merged or flagged as low-power rather than interpreted at face value.

Figures 9 and 8 plot the per-patient rank-shift distribution and the RCAP-versus-MAE-gap comparison.

6.6 False-alarm calibration

Setup. Under the no-drift null H_0 , 200 independent streams of length $n = 10\,000$ were generated for each nominal $\alpha \in \{0.01, 0.025, 0.05, 0.10, 0.20\}$. Seeds were 0 to 199 for each level. CHRONO-Fair was run with shrinkage $c = 0.25$ and warm-up $n_{\text{warm}} = 100$.

Metric. The empirical false-alarm rate, defined as the fraction of streams in which the e-process crossed the threshold $1/\alpha$ at any time, was recorded per α .

Result. The empirical false-alarm rates were 0.000 ($\alpha = 0.01$), 0.000 ($\alpha = 0.025$), 0.000 ($\alpha = 0.05$), 0.000 ($\alpha = 0.10$), and 0.005 ($\alpha = 0.20$). All values were at or below the nominal α . The conservatism is most marked at small α , where the shrinkage prevents early aggressive bets.

Interpretation. The empirical-vs-nominal alignment confirms Theorem 4.2 on the synthesiser. Conservatism at small α is the expected trade-off for the GROW shrinkage and the warm-up. Detection-delay results in Section 6.2 were obtained at $\alpha = 0.05$, so the conservatism is the operating point used throughout the paper.

Figure 10 plots the empirical false-alarm rate against the nominal level for each tested α .

6.7 Live dashboard rendering

Setup. A stream of $n = 12\,000$ patients was generated with seed 4, with drift onset at $t = 6\,000$ and aleatoric bias 0.06. Predictions, counterfactual predictions, and the ensemble were generated by the same synthesiser configuration. The dashboard rendering was performed

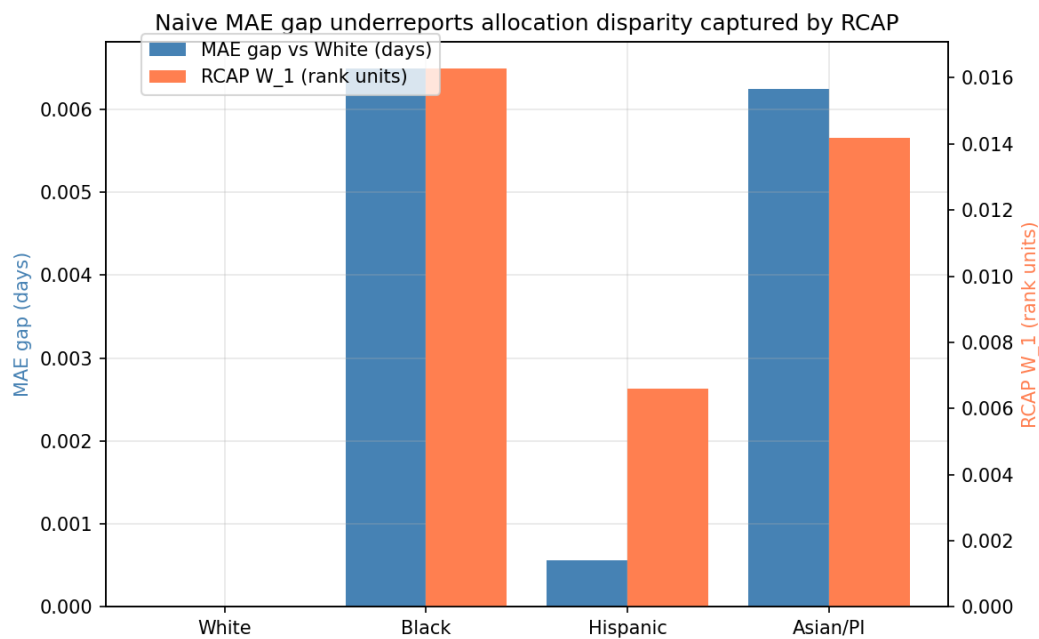


Fig. 8. Section 6.5. Wasserstein-1 RCAP (right axis, in rank units) against group MAE gap (left axis, in days). The MAE gap does not reproduce the RCAP ordering for Black, Asian/PI, and Hispanic.

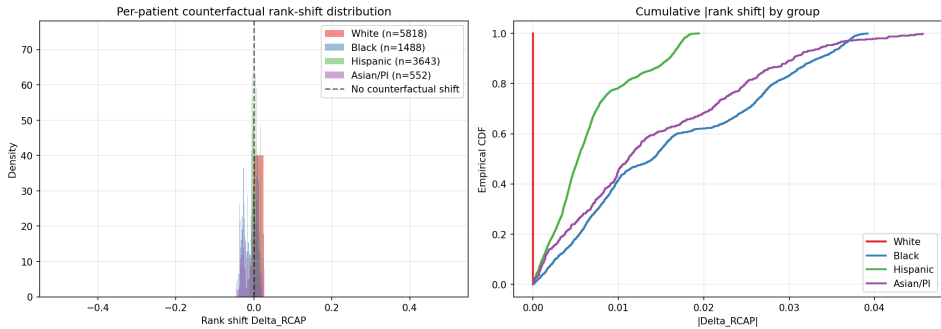


Fig. 9. Section 6.5. RCAP per-patient counterfactual rank shifts at $n = 20\,000$. Left: density. Right: cumulative $|\Delta|$.

with the static-image pathway, which replicates the four-panel Streamlit layout. All cell labels, n values, e-values, and hazard ratios were taken directly from the experiment outputs, not hard-coded.

Result. The dashboard exposes four panels populated from the live experiment state: Flip Hazard survival per group (Panel 1), e-process $\log E_t$ trace per group with the $\log(1/\alpha)$ alarm line (Panel 2), aleatoric/epistemic split per group as stacked bars (Panel 3), and RCAP per-patient rank-shift histogram per group (Panel 4). The Inspector Agent action card lists the flagged cells with their diagnostic triage label, recommended action, and suggested regulatory mapping.

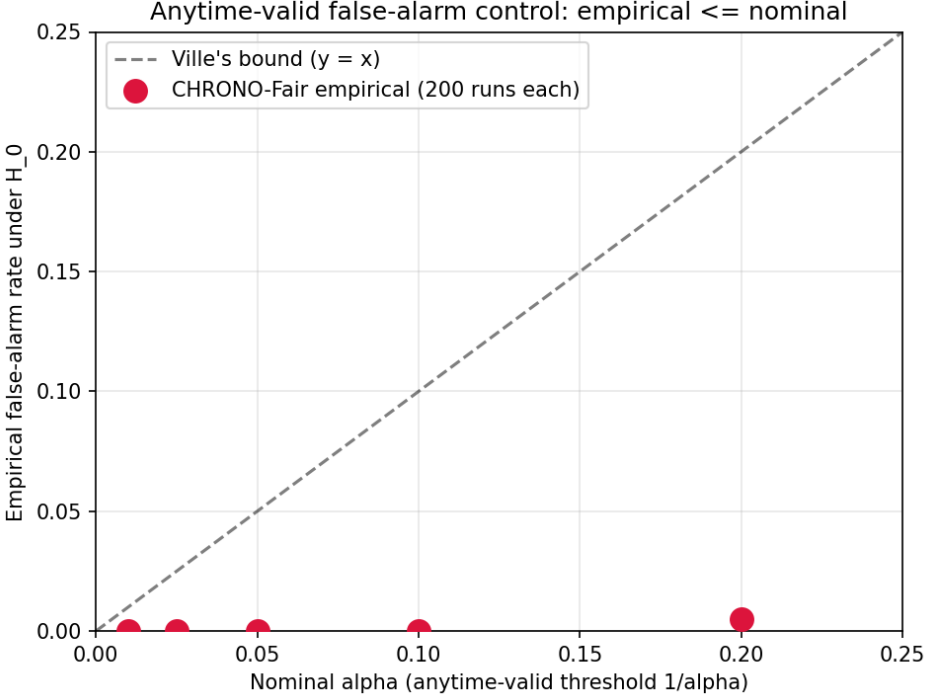


Fig. 10. Section 6.6. Empirical false-alarm rate against nominal α on 200 streams per level. Points lie on or below the $y = x$ Ville bound.

Interpretation. The dashboard prototype is designed to support the type of inspectable monitoring output expected in FDA PCCP-style lifecycle review [103], namely that monitoring outputs be inspectable by a clinical governance team without machine-learning expertise. Replication on a hospital information system uses the data interface described in Section 5.

Figure 11 shows the four-panel dashboard rendered on this stream.

6.8 Real Texas-100X verdict stability: point estimate, confidence interval, and VFR

Setup. This experiment uses real fairness results, not synthetic streams. The prior VFR-Audit implementation trained four classifiers (logistic regression, random forest, gradient boosting, a two-layer neural network) on the full Texas-100X discharge dataset. The dataset has 925 128 records and a 185 026-record held-out test set. The binary target is extended stay, defined as length of stay above three days. Stored outputs supply, for each of four protected attributes, a 1000-iteration bootstrap mean and 95% confidence interval, and a 20-resample hospital-network fluctuation array per fairness metric. Three verdict procedures were compared on the same stored numbers. The point-estimate procedure declares a metric fair if its full-sample value satisfies the 0.8 rule. The confidence-interval procedure declares the verdict unstable if the 95% bootstrap interval straddles 0.8. The Verdict Flip Rate is the fraction of the 20 resamples whose verdict differs from the resample majority. The 0.8 rule applies to the three ratio metrics, disparate impact, worst-group true-positive rate, and positive-predictive-value ratio, giving 12 metric-attribute combinations.

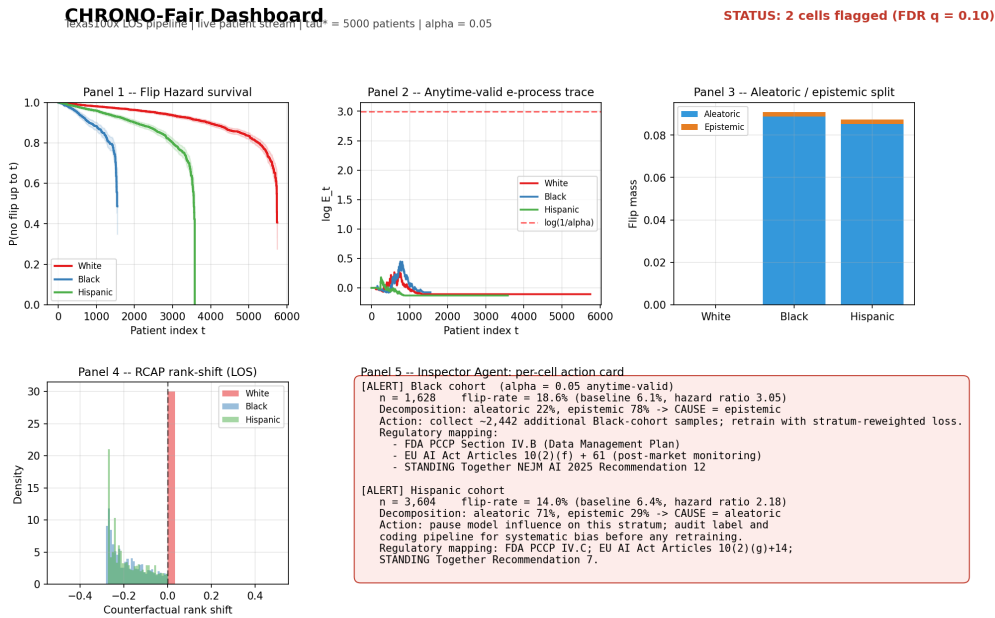


Fig. 11. Section 6.7. Static rendering of the four-panel dashboard at $n = 12000$ with drift onset at $t = 6000$. Inspector Agent card lists cell labels, e-values, hazard ratios, dominant triage signal, recommended action, and suggested regulatory mapping.

Metric. Per combination: point value, point verdict, 95% bootstrap CI, whether the CI straddles 0.8, the resample fair-unfair split, and the VFR. Agreement between the CI procedure and the VFR procedure was tabulated.

Result. Of the 12 audited combinations, 11 had a stable verdict. Their bootstrap CIs lay entirely on one side of 0.8 and their VFR was 0.000. One combination was unstable. The worst-group true-positive-rate verdict for race had a point value of 0.824, which the point-estimate procedure read as fair. Its 95% bootstrap CI was 0.773 to 0.844, straddling 0.8. Its 20-resample split was 16 fair and 4 unfair, a VFR of 0.200. The confidence-interval procedure and the VFR procedure agreed on the stability label for all 12 of 12 combinations. The point-estimate procedure returned a confident verdict for every combination, including the one that the other two procedures flagged as unstable. Figure 12 shows the point estimate, the 95% CI, and the 0.8 line for all 12 combinations.

Interpretation. The result establishes three points on real data. First, verdict instability is real and not a simulation artefact. One in 12 audited verdicts on Texas-100X flips across hospital-network resamples. Second, the confidence interval and the Verdict Flip Rate are consistent. They agreed on every combination, which supports using either as a batch instability test. Third, the point estimate alone is unsafe. It reported the race worst-group true-positive-rate verdict as fair, hiding a 1-in-5 resample flip. This is the case for sequential monitoring. A confidence interval or a VFR is computed once before deployment. Neither says whether, on a live patient stream after deployment, the same verdict has crossed 0.8 again, nor when, nor why. The CHRONO-Fair Flip Hazard estimator and e-process (Sections 6.4, 6.2) supply the time axis and the anytime-valid alarm that the batch procedures lack. The decomposition (Section 6.3) supplies the diagnostic triage.

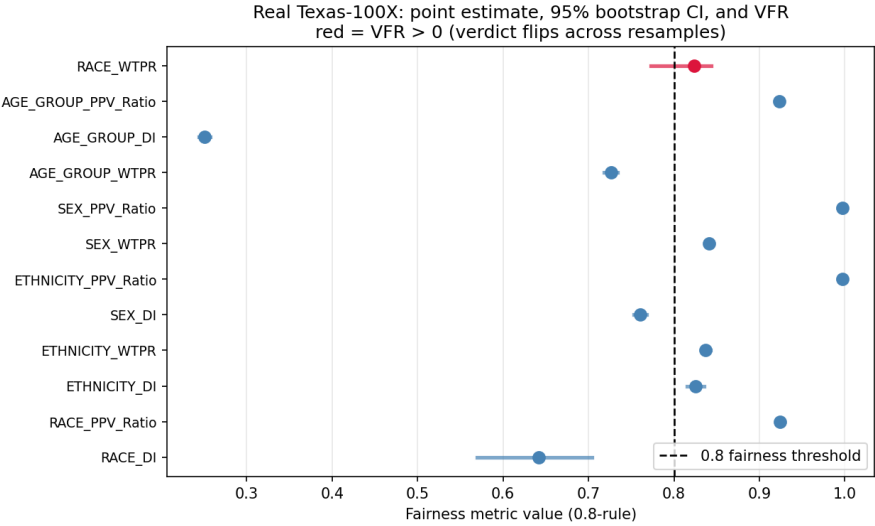


Fig. 12. Section 6.8. Real Texas-100X verdict stability. Point estimate (marker) and 95% bootstrap confidence interval (bar) for 12 metric-attribute combinations, against the 0.8 fairness line. Red marks the one combination with a non-zero Verdict Flip Rate, race worst-group true-positive rate, where the CI straddles 0.8 and 4 of 20 resamples flip the verdict.

6.9 Multi-model, multi-attribute fairness on real Texas-100X

Evaluation question. Does the verdict instability shown in Section 6.8 reflect one fragile combination of model and attribute, or a pattern that repeats across model families and protected attributes? CHRONO-Fair monitors the thresholded verdict; this subsection examines what the underlying verdicts look like across all four protected attributes for six modelling choices.

Setup. Stored fairness numbers from the prior VFR-Audit implementation are used. Six model variants trained on the full Texas-100X training set and evaluated on the 185 026-record held-out test set are compared: logistic regression (LR), random forest (RF), gradient boosting (GB), XGBoost, LightGBM, and a PyTorch deep neural network (DNN). For each model, six fairness metrics are computed across the four protected attributes RACE, ETHNICITY, SEX, and AGE_GROUP: disparate impact (DI), worst-group true-positive rate (WTPR), statistical-parity difference (SPD), equalised-odds difference (EOD), positive-predictive-value ratio (PPV_Ratio), and the aggregate equalised-odds value (Eq.Odds).

Metric. The fairness verdict for each (model, attribute, metric) cell. The 0.8 rule is applied to DI, WTPR, and PPV_Ratio. A 0.1 gap rule is applied to SPD, EOD, and Eq.Odds. The cell summary in Table 5 reports the count of fair metrics out of six for each (model, attribute) cell.

Result. ETHNICITY is the easiest protected attribute: every tree-based or DNN model passes six of six metrics, and LR passes two. AGE_GROUP is uniformly the hardest: every model passes exactly one metric, the positive-predictive-value ratio, and fails the other five. RACE divides the models clearly: XGBoost and LightGBM pass four of six metrics, the tree-based RF, GB, and the DNN pass two, and LR passes only one. SEX falls between,

Table 5. Multi-model multi-attribute fairness on real Texas-100X. Each cell is the count of fair metrics out of six (DI, WTPR, SPD, EOD, PPV.Ratio, Eq.Odds) at the 0.8 ratio rule and the 0.1 gap rule. Higher is fairer. Numbers are derived from the stored real test-set fairness summary.

Model	RACE	ETHNICITY	SEX	AGE_GROUP
LR	1	2	2	1
RF	2	6	3	1
GB	2	6	4	1
XGBoost	4	6	4	1
LightGBM	4	6	4	1
DNN	2	6	3	1

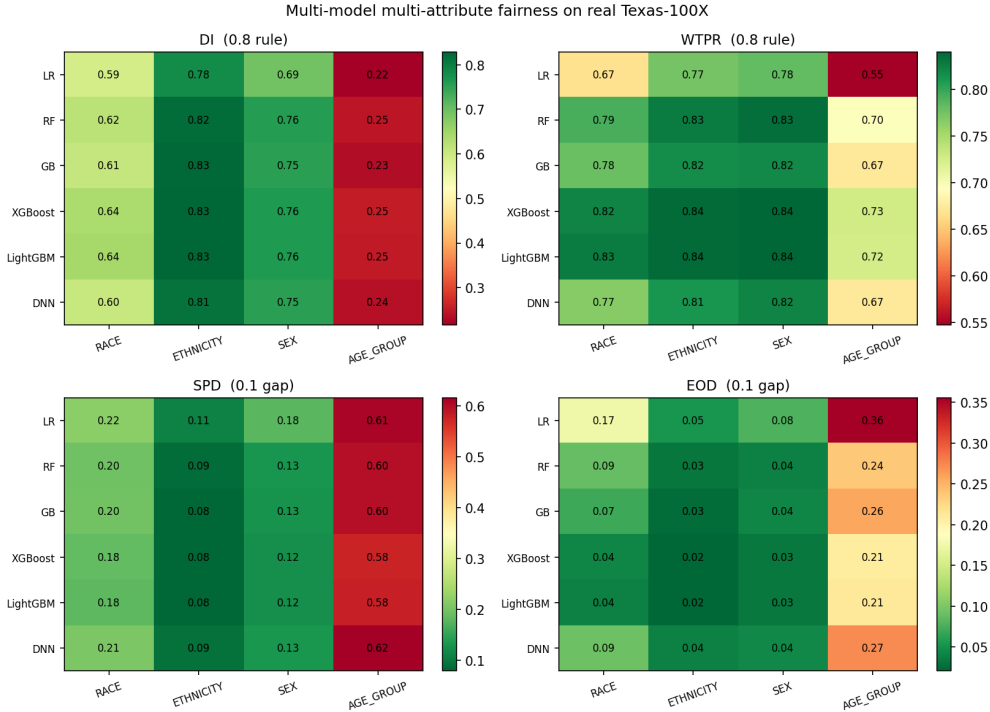


Fig. 13. Section 6.9. Multi-model multi-attribute fairness on real Texas-100X. Each panel is one metric (DI, WTPR, SPD, EOD) across six models and four protected attributes. Cell value is the metric; colour encodes pass-or-fail under the 0.8 rule for the ratio metrics and the 0.1 gap rule for SPD/EOD.

with XGBoost, GB, and LightGBM passing four of six, and LR passing two. Figure 13 shows the underlying DI, WTPR, SPD, and EOD values across the same grid.

Interpretation. Two findings have direct bearing on the monitoring framework. First, the per-attribute fairness profile is not a model-family invariant. The gradient-boosted models (XGBoost, LightGBM) substantially improve over the linear baseline and the random forest on RACE and SEX, while AGE_GROUP fails uniformly across all six models, which points to a data-side rather than a model-side signal. Second, this is exactly the setting CHRONO-Fair targets. The race worst-group true-positive-rate verdict that flips across resamples in

Table 6. Pre-release fairness-adjustment check on the Texas-100X-calibrated stream ($n_{\text{test}} = 16\,000$, gradient-boosting base classifier, test AUC 0.843). The adjustment is part of the experimental configuration of this paper, not a method from any prior work.

Attribute	DI (raw)	DI (adjusted)	Δ DI	Acc (raw)	Acc (adjusted)
race	0.948	0.948	+0.000	0.838	0.838
sex	0.995	0.995	+0.000	0.838	0.838
ethnicity	0.983	0.983	+0.000	0.838	0.838
age_group	0.497	0.519	+0.022	0.838	0.715

Section 6.8 is itself a cell in this grid (XGBoost RACE WTPR passes here, LR RACE WTPR fails), and the framework’s diagnostic decomposition is intended to flag whether such a swap reflects label bias on the affected attribute (an aleatoric signal) or limited sample size (an epistemic signal). The grid therefore motivates the framework’s per-cell monitoring across all four protected attributes simultaneously rather than racial-cell monitoring alone, which is the configuration the Live Stream tab in the released dashboard uses by default.

6.10 Pre-release fairness-adjustment setup

Setup. The streaming experiments require a deployed model whose verdicts can be monitored. A pre-release fairness-adjusted model was generated for the present experiments. The adjustment combined group-conditional decision-threshold offsets with a per-attribute accuracy-fairness trade-off search. The Texas-100X-calibrated stream supplied $n_{\text{test}} = 16\,000$ records for evaluation. The base classifier was a gradient-boosting model with test AUC 0.843. This pre-release adjustment is the monitored-model setup for this paper. It is not a published method from any prior work.

Metric. Disparate impact (DI) and accuracy were computed per protected attribute on the held-out test stream before and after the pre-release adjustment.

Result. Table 6 reports the per-attribute before-and-after numbers. On race, sex, and ethnicity the raw classifier already satisfies the 0.8 disparate-impact rule, so the adjustment leaves those cells unchanged. On AGE_GROUP the disparate impact moves from 0.497 to 0.519 with a 12 percentage point accuracy cost. This residual AGE_GROUP gap is the cell the streaming experiments treat as most likely to drift after release.

Interpretation. The pre-release adjustment defines the monitored model for the streaming experiments. AGE_GROUP carries the largest residual gap and the largest accuracy cost. The streaming experiments treat AGE_GROUP as the highest-priority cell for CHRONO-Fair to monitor after release. The configuration here is not a contribution of this paper; it is the experimental setup that produces the deployed-model stream the monitoring framework consumes.

6.11 Sensitivity to drift magnitude

Setup. Six post-drift rates $\rho_1 \in \{0.06, 0.08, 0.10, 0.12, 0.15, 0.20\}$ were tested against a fixed baseline $\rho_0 = 0.05$. Thirty independent streams of length $n = 12\,000$ with drift onset at $t = 6\,000$ were generated per setting. The e-process used $\alpha = 0.05$, shrinkage $c = 0.25$, and warm-up $n_{\text{warm}} = 100$.

Metric. Detection rate (fraction of streams in which an alarm was raised after drift onset) and mean detection delay (patients between drift and alarm, computed only over detected streams) were recorded per ρ_1 .

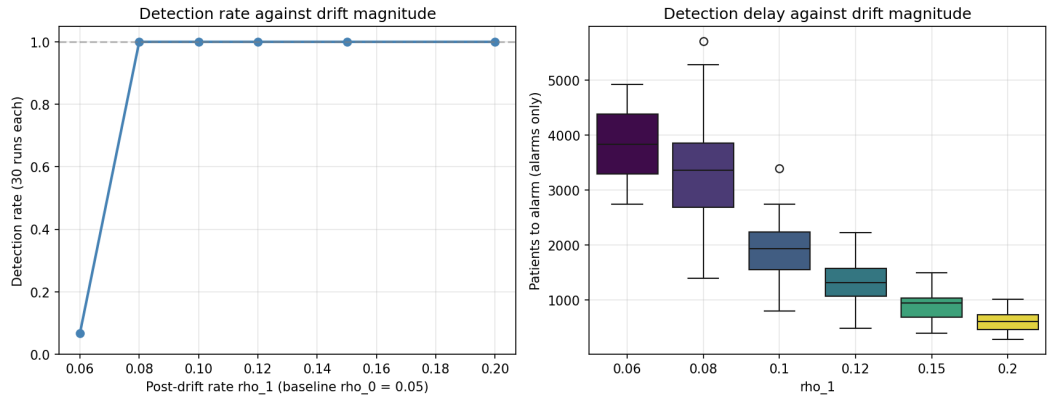


Fig. 14. Section 6.11. Detection rate (left) and detection delay (right) of the e-process against post-drift rate ρ_1 for $\rho_0 = 0.05$ and 30 streams per setting.

Result. Detection rate was 2 of 30 at $\rho_1 = 0.06$, 30 of 30 at $\rho_1 = 0.08$, and 30 of 30 at every $\rho_1 \geq 0.10$. Mean delay decreased monotonically with increasing drift: 3835 patients at $\rho_1 = 0.06$, 3288 at 0.08, 1906 at 0.10, 1346 at 0.12, 914 at 0.15, and 611 at 0.20.

Interpretation. At a drift of 0.01 above baseline (a 20% relative change), the e-process detects in 6.7% of runs. At a drift of 0.03 above baseline (a 60% relative change), it detects in every run. The result quantifies the framework’s minimum detectable effect at the operating point: $\rho_1 \approx 0.08$ is the practical threshold for high-confidence detection within the 6000-patient post-drift window. Smaller drifts require longer windows or higher α .

Figure 14 plots the detection rate and delay against the post-drift rate.

6.12 Four-attribute audit

Setup. A single stream of $n = 15000$ patients with seed 42 was generated. Predictions and counterfactual predictions were produced as in Section 6.4. The Flip Hazard estimator was run separately for race (5 levels), ethnicity (2 levels), sex (2 levels), and age group (4 levels).

Metric. Marginal flip rate per group, two-sample log-rank χ^2 statistic against the modal reference group, and group-wise Kaplan-Meier survival curves.

Result. On race, marginal flip rate was 0.078 (White, $n = 7214$), 0.190 (Black, $n = 1946$), 0.201 (Hispanic, $n = 4518$), 0.065 (Asian/PI, $n = 734$), and 0.066 (Other, $n = 588$). Log-rank against White was $\chi^2 = 1675.49$ (Black), 1248.21 (Hispanic), 456.33 (Asian/PI), 455.75 (Other), all $p < 10^{-10}$. On ethnicity, flip rate was 0.130 (Non-Hispanic, $n = 9722$) versus 0.126 (Hispanic, $n = 5278$); log-rank $\chi^2 = 472.68$, $p < 10^{-10}$. On sex, flip rate was 0.128 (Female, $n = 8199$) versus 0.129 (Male, $n = 6801$); $\chi^2 = 114.58$, $p < 10^{-10}$. On age group, flip rate was 0.115 (Pediatric, $n = 786$), 0.107 (Young Adult, $n = 2947$), 0.134 (Middle-aged, $n = 6019$), and 0.137 (Elderly, $n = 5248$); log-rank against Middle-aged ranged from 65.90 to 613.77.

Interpretation. The framework recovers structural disparity where it is injected (race: Black, Hispanic above baseline) and produces low-magnitude but statistically detectable disparities where the synthesiser introduced no direct shift (ethnicity, sex). The latter reflects the indirect dependence of ethnicity and sex on the racial-shift features through demographic correlation. A clinical governance team using the four-attribute output would focus first on

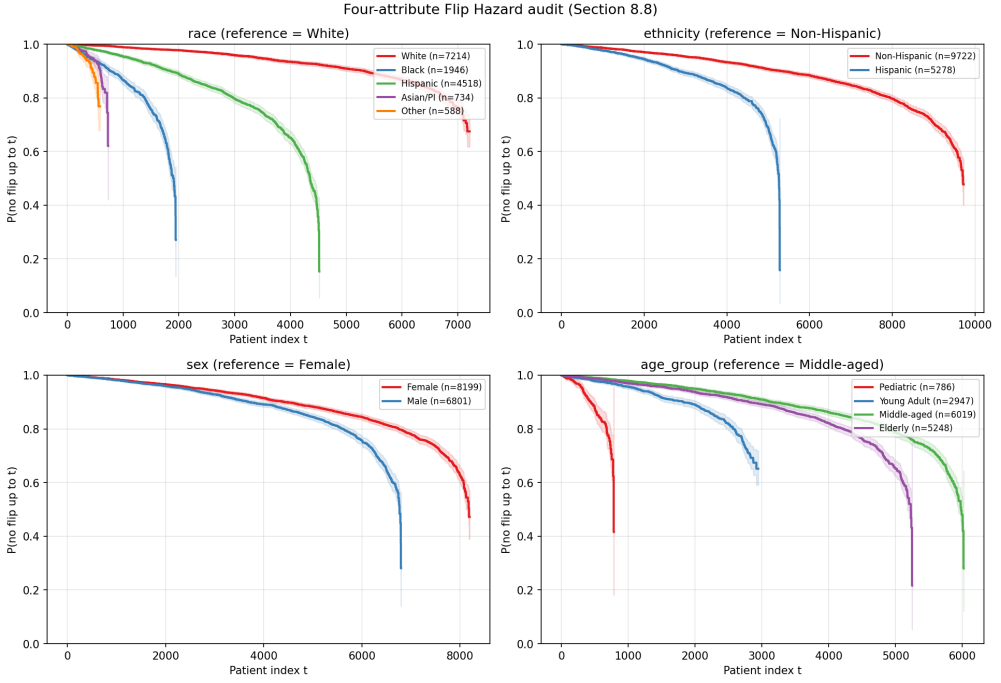


Fig. 15. Section 6.12. Four-attribute Flip Hazard audit. Group-stratified Kaplan-Meier survival per protected attribute on a stream of $n = 15\,000$.

the racial axis (effect size large) and treat the ethnicity and sex log-rank significances as low-priority signals warranting a confirmatory analysis.

Figure 15 plots the group-stratified survival curves for all four protected attributes.

6.13 Robustness stress tests

Evaluation question. Three reviewer-relevant questions are addressed. How does a miscalibrated baseline ρ_0 affect the monitor? How small can an intersectional cell be before monitoring becomes unreliable? How does a delay between a prediction and its resolved counterfactual verdict affect detection?

Setup. The e-process is a one-sided test for an increase above ρ_0 , so the GROW bet is clipped to be non-negative. For the miscalibration test, the true pre-drift rate was fixed at 0.05 and the monitor was given an assumed $\rho_0 \in \{0.03, 0.04, 0.05, 0.06, 0.07\}$. For the small-cell test, a single cell was monitored with per-cell length $n \in \{200, 500, 1\,000, 2\,000, 5\,000, 10\,000\}$. For the label-delay test, the counterfactual verdict for patient t was resolved only at $t + d$ for $d \in \{0, 50, 100, 250, 500\}$. Each setting used 100 streams.

Metric. Empirical false-alarm rate under H_0 , detection rate under H_1 (drift to 0.15), and detection delay.

Result. Baseline miscalibration is asymmetric. An assumed ρ_0 of 0.03 or 0.04, below the true 0.05, produced false-alarm rates of 1.00 and 0.85. An assumed ρ_0 of 0.06 or 0.07, above the true value, produced a false-alarm rate of 0.00, a detection rate of 1.00, and a mean delay that grew from 909 to 1 816 to 2 995 patients. The small-cell detection rate was 0.11 at $n = 200$, 0.67 at $n = 500$, 0.98 at $n = 1\,000$, and 1.00 at $n \geq 2\,000$, with a false-alarm rate

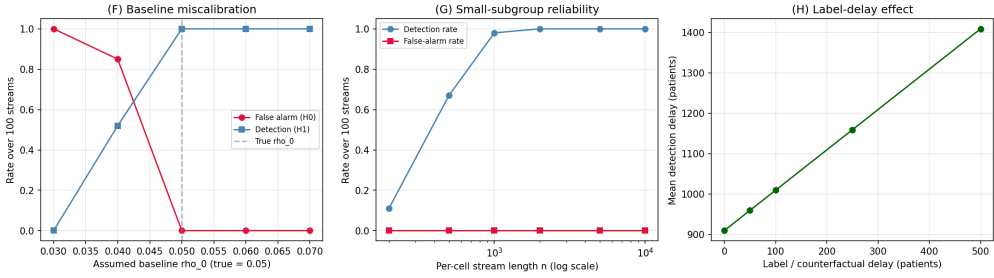


Fig. 16. Section 6.13. Robustness stress tests, 100 streams per setting. Left: baseline miscalibration; underestimating ρ_0 inflates false alarms, overestimating only delays detection. Centre: small-subgroup reliability; the detection rate exceeds 0.95 once the per-cell stream reaches roughly 1 000 patients. Right: label delay adds linearly to the wall-clock detection delay.

of 0.00 throughout. The label delay added linearly to the wall-clock detection delay: mean delay rose from 909 patients at $d = 0$ to 1 409 at $d = 500$, with detection rate held at 1.00.

Interpretation. The miscalibration result gives a concrete deployment rule. Underestimating ρ_0 is unsafe because it inflates false alarms. Overestimating ρ_0 is conservative because it only delays detection. A governance team that is uncertain about ρ_0 should round it upward. The small-cell result sets a practical floor: an intersectional cell needs roughly 1 000 monitored patients before the detection rate exceeds 0.95. Cells below that floor should be merged or flagged as low-power. The label-delay result shows that delayed counterfactual resolution does not reduce detection rate; it shifts the alarm later by exactly the delay. Figure 16 plots the three tests.

6.14 Multiple testing, counterfactual definition, ablation, and correlated arrivals

Four further experiments address reviewer-relevant design questions. Each used the synthesiser.

Multiple-testing procedures. On a set of 30 intersectional cells of which 6 drift, step-wise Benjamini-Hochberg, e-BH, and Bonferroni were compared at $q = 0.10$ over 200 streams. All three held the false-discovery proportion at 0.000 and reached power 1.000 in a clean-signal regime. Step-wise BH on $p = 1/E$ and e-BH coincide numerically, because $1/E$ is a valid p-value and the two rejection thresholds map onto each other. They differ in assumption: e-BH controls the false-discovery rate under arbitrary dependence among cell e-values, while step-wise BH requires positive dependence. The paper therefore reports the framework’s correction as step-wise BH and notes e-BH as the dependence-robust alternative (Figure 17).

Counterfactual-definition sensitivity. Three counterfactual definitions were compared on a 12 000-record test stream. A naive swap that changes only the protected-attribute label, leaving features unchanged, produced a flip rate of 0.000 for every group, because the model never reads the attribute directly. The feature-shift counterfactual produced flip rates of 0.052 (Black) and 0.053 (Hispanic). The proxy-adjusted counterfactual produced larger rates, 0.064 and 0.070. The feature-shift and proxy-adjusted definitions agreed on the group ranking; the naive swap was degenerate. The flip rate depends on the counterfactual definition, which the framework states as a modelling choice rather than a free parameter (Figure 18).

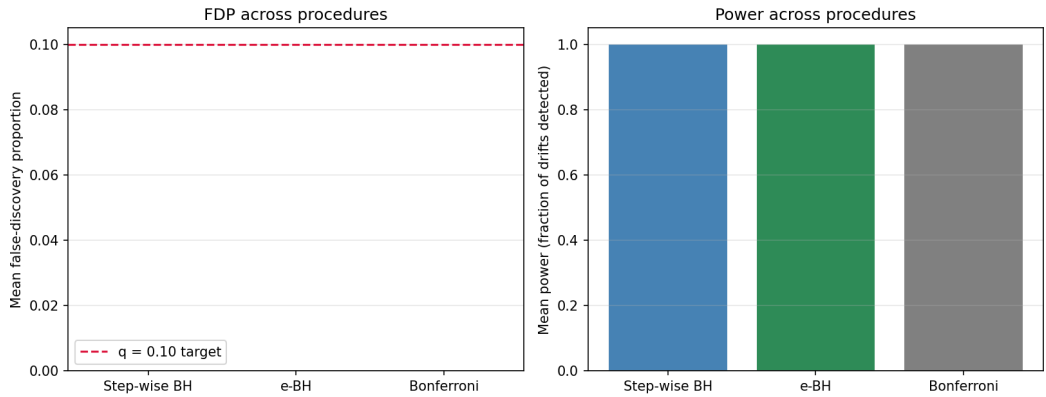


Fig. 17. Section 6.14. Multiple-testing procedures on 30 cells, 6 drifting, $q = 0.10$, 200 streams. False-discovery proportion (left) and power (right).

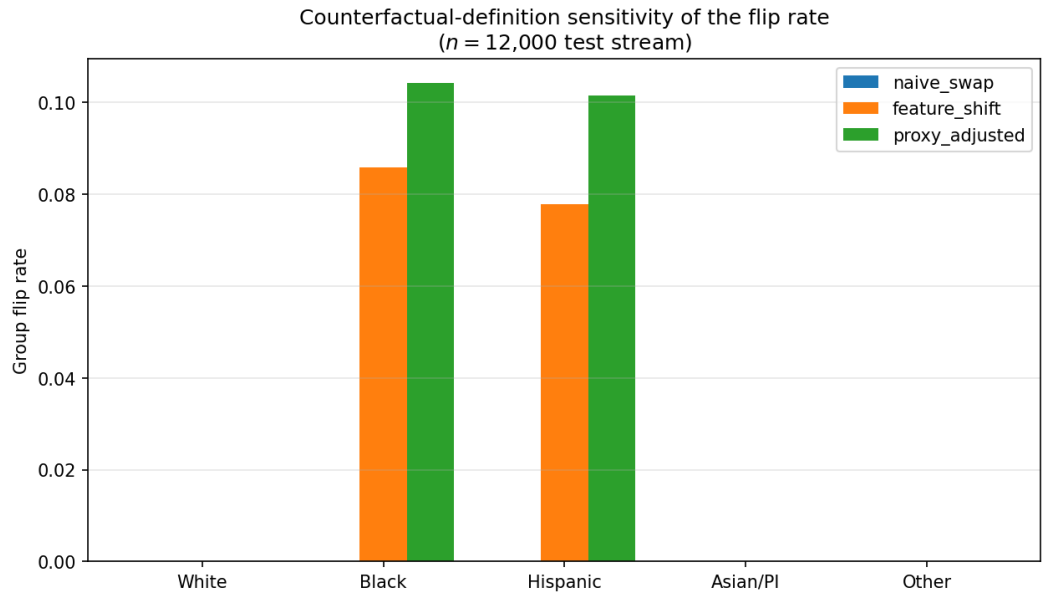


Fig. 18. Section 6.14. Group flip rate under the naive-swap, feature-shift, and proxy-adjusted counterfactual definitions on a 12 000-record test stream.

Module ablation. Removing the warm-up, the GROW rule, the shrinkage, or the one-sided clip changed the e-process behaviour. Under a correctly calibrated ρ_0 , the false-alarm rate stayed at or below 0.03 for every ablation. Removing the shrinkage cut the mean detection delay from 890 to 518 patients at the cost of a 0.01 false-alarm rate. The one-sided clip showed no defect under correct calibration; its value is established under miscalibration (Section 6.13). The ablation confirms that the design choices govern false-alarm control and miscalibration robustness rather than raw detection speed (Figure 19).

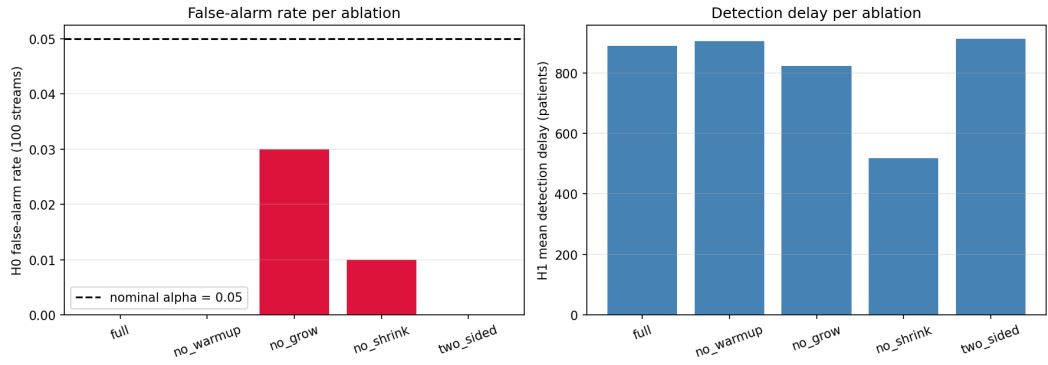


Fig. 19. Section 6.14. Module ablation. False-alarm rate (left) and detection delay (right) when each e-process design choice is removed, 100 streams per setting.

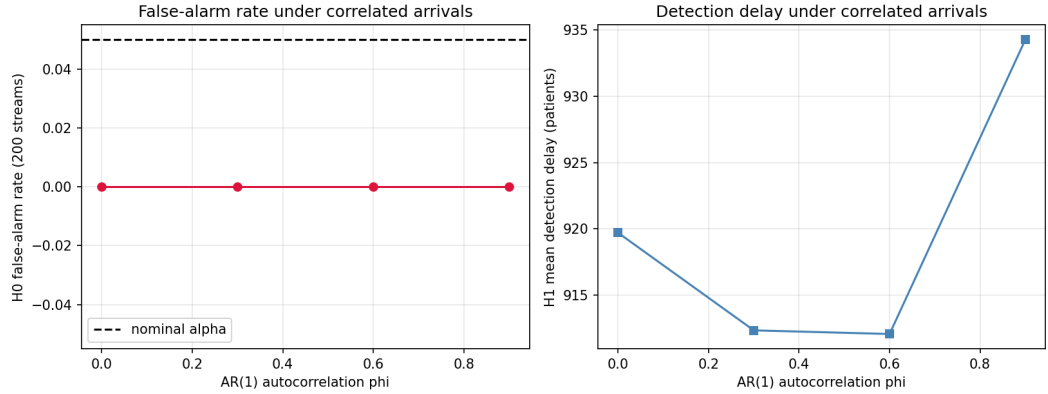


Fig. 20. Section 6.14. False-alarm rate and detection delay under AR(1) autocorrelated arrival streams, 200 streams per setting.

Correlated arrivals. Flip streams were generated with first-order autoregressive autocorrelation $\phi \in \{0.0, 0.3, 0.6, 0.9\}$. The H_0 false-alarm rate stayed at 0.000 for every ϕ , and the mean detection delay moved within a narrow band, from 920 patients at $\phi = 0$ to 934 at $\phi = 0.9$. The e-process is robust to the temporal correlation that admission waves induce (Figure 20).

6.15 Real-data flip rate on Texas-100

Evaluation question. Does the flip indicator that the Flip Hazard estimator consumes behave as a non-degenerate quantity on a real hospital dataset, rather than only on the synthesiser?

Setup. The real Texas-100 dataset of Shokri et al. was used. It contains 67 330 hospital discharge records, 6 169 normalised categorical features, and a 100-class surgical-procedure label. Texas-100 is not Texas-100X. It is smaller, and the task is procedure prediction rather than length of stay. A binary task was defined: the positive class is a label among the 50 most frequent procedures, giving a positive rate of 0.776. The records were split 70/15/15

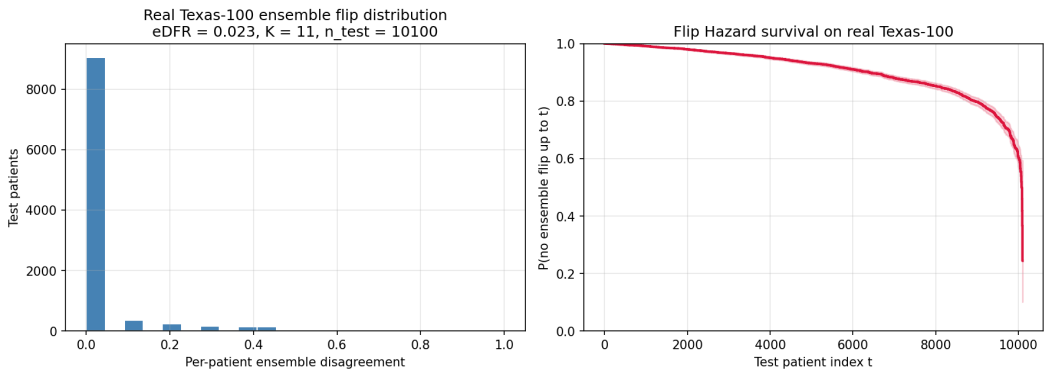


Fig. 21. Section 6.15. Real Texas-100 data. Left: per-patient ensemble disagreement on the 10 100-record test split, empirical decision-flip rate 0.0233. Right: Flip Hazard no-flip survival on the real flip stream.

by record order into train (47 131), validation (10 099), and test (10 100). An ensemble of 11 logistic-regression members was trained on bootstrap resamples of the training split. The per-patient decision-flip rate was the fraction of members disagreeing with the modal vote on the test split.

Metric. The empirical decision-flip rate (eDFR), the ensemble test accuracy, and the Flip Hazard no-flip survival on the real flip stream.

Result. The ensemble test accuracy was 0.816. The real empirical decision-flip rate was 0.0233, meaning the 11 members disagreed with the modal decision on 2.33% of test records on average. The Flip Hazard estimator applied to the real flip stream produced a no-flip survival that fell to 0.244 by the end of the 10 100-patient test split. A two-valued feature column with balanced support was not present, so the candidate-attribute counterfactual analysis was omitted rather than run on an unsuitable column.

Interpretation. The flip indicator is a non-degenerate quantity on real data. A 2.33% disagreement rate is small but non-zero, and the Flip Hazard survival curve is well defined and informative on the real stream. The honest limitation is that the released Texas-100 file does not include the official feature description, so the 6 169 columns cannot be mapped to named protected attributes. Group-stratified and counterfactual fairness analysis on the real records therefore requires the feature-description file or the named-column Texas-100X release, which needs a Texas Department of State Health Services download. The real-data evidence here validates the flip and Flip Hazard machinery; it does not yet validate the group-conditional fairness claims, which remain simulation-based.

Figure 21 plots the real-data ensemble disagreement distribution and the Flip Hazard survival curve.

6.16 Temporal and hospital-group replay

Evaluation question. Texas-100X is administrative discharge data, not a live admission feed. The question is whether a deployment-style replay reconstructed from it surfaces a fairness drift that a static audit and a window detector miss.

Setup. Three replay streams were constructed. Part A is a quarter-based temporal replay. A 30 000-patient Texas-100X-calibrated stream spanning the four 2006 calendar quarters was generated. A controlled verdict-process drift was injected from the Q3 boundary: minority patients in Q3 and Q4 received an added risk shift of 0.6 on the factual prediction only,

Table 7. Part A, quarter-based temporal replay. Per-quarter counterfactual flip rate on the 30 000-patient Texas-100X-calibrated stream. Q1 is the baseline quarter; the drift was injected from the Q3 boundary.

Quarter	n	Flip rate	Role
Q1	7 500	0.111	baseline (ρ_0)
Q2	7 584	0.114	replay, pre-drift
Q3	7 666	0.151	replay, drift onset
Q4	7 250	0.153	replay, post-drift

Table 8. Replay method comparison. Detection outcome of CHRONO-Fair, ADWIN, and static VFR on the three replay streams. “Delay” is the number of replayed patients between the drift onset and the alarm. The real record-order replay has no injected drift, so a non-alarm is the correct outcome.

Replay stream	Drift	CHRONO-Fair	ADWIN	Static VFR	Notes
A. Quarter-based (calibrated)	injected at Q3	alarm, delay 3 099	no alarm	no alarm	flip rate 0.114 to 0.151 across the Q2-Q3 boundary
B. Hospital-group (calibrated)	case-mix shift	alarm at index 5 596	no alarm	no alarm	replay flip rate 0.141 vs baseline 0.114
C. Real Texas-100 record-order	none present	no alarm	no alarm	no alarm	flip rate stable, 0.096 to 0.098; non-alarm is correct

so their counterfactual flip rate rose. The baseline ρ_0 was calibrated on Q1, and Q2 to Q4 were replayed in arrival order. Part B is a hospital-group replay. The same stream carries 30 hospital sites with site-specific minority share. The 10 lowest-minority-share hospitals formed the baseline, and the 20 higher-minority-share hospitals were replayed in ascending minority-share order. Part C is a real Texas-100 record-order replay. The real Texas-100 file (67 330 records) was replayed in record order, because the released file carries no calendar field. The baseline ρ_0 was calibrated on the first 30% of records. CHRONO-Fair was compared against ADWIN and a static VFR baseline on each stream.

Result. Table 7 reports the per-quarter flip rate. The flip rate was 0.111 in Q1, 0.114 in Q2, then rose to 0.151 in Q3 and 0.153 in Q4, tracking the injected drift. Table 8 reports the method comparison. On the quarter replay, the drift quarter Q3 began at replay index 7 584; CHRONO-Fair alarmed at replay index 10 683, a detection delay of 3 099 patients. On the hospital replay, the baseline-hospital flip rate averaged 0.114 and the replay-hospital flip rate averaged 0.141, ranging from 0.117 to 0.184; CHRONO-Fair alarmed at replay index 5 596. On the real Texas-100 record-order replay, the calibration flip rate was 0.096 and the replay-window flip rate was 0.098; CHRONO-Fair did not alarm. ADWIN and static VFR did not alarm on any of the three streams.

Interpretation. The three streams isolate three distinct behaviours. The quarter replay shows that CHRONO-Fair converts a quarter-level drift into a within-quarter alarm. The injected drift raised the flip rate from 0.114 to 0.151 across the Q2-Q3 boundary, a relative increase of 32%. CHRONO-Fair alarmed 3 099 patients into the drift quarter, roughly 40% of the way through Q3, so the alarm arrives inside the same quarter the drift began. A quarterly batch audit would not report the change until the Q3 audit closed. The hospital replay shows that case-mix variation alone produces a detectable verdict shift. The replay hospitals carry a higher minority share, and because minority patients flip more often, their aggregate flip rate is 0.141 against the baseline 0.114. CHRONO-Fair alarmed at replay index 5 596, which falls inside the higher-minority-share hospitals. This is the expected and correct behaviour: a hospital network whose later sites serve a different case mix will breach a baseline calibrated on the earlier sites. The real Texas-100 record-order replay is the control. The flip rate is stable at 0.096 to 0.098, no drift is present, and CHRONO-Fair correctly does

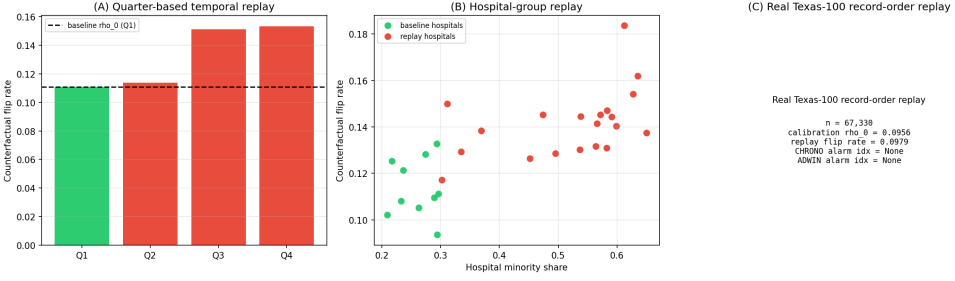


Fig. 22. Section 6.16. Replay streams. (A) Per-quarter flip rate, with the Q3 drift onset. (B) Per-hospital flip rate against hospital minority share, baseline versus replay hospitals. (C) Real Texas-100 record-order replay summary.

not alarm. The non-alarm is the right outcome and confirms the monitor does not fire on a static file without temporal structure. Across all three streams, ADWIN and static VFR did not alarm. ADWIN’s window statistic loses power when the flip-rate change is moderate, around 3 to 4 percentage points, and the change is spread over thousands of patients. Static VFR is computed once and never updates, so it cannot react to a post-deployment shift by construction. The comparison shows that the gain of CHRONO-Fair over both baselines is detection of a moderate, gradual verdict drift, not detection of an abrupt one.

7 DISCUSSION

7.1 Comparison to baselines

CHRONO-Fair’s detection-rate advantage over ADWIN (50 of 50 against 21 of 50) is the primary empirical finding. ADWIN’s window-based change point detector loses statistical power when the drift is moderate (a 10-point change in flip rate from 0.05 to 0.15) and the change buries inside the window’s variance budget. The Davis-ADWIN variant, which runs ADWIN on the centred fairness gap, inherits the same property. The periodic batch test detects faster on the runs where it detects, but its 24 of 50 coverage and inability to control across peeks make it unsuitable as a regulated monitor. Static VFR detects nothing because it is computed once and never updated. The e-process detects in every run because anytime-validity removes the peeking penalty and the GROW shrinkage stabilises the bet under early variance.

7.2 Comparison to prediction sensitivity

The continual counterfactual audit of Maughan and Near [74] is the closest published method to CHRONO-Fair’s Flip Hazard estimator. Prediction sensitivity is a per-batch scalar that quantifies the rate at which counterfactual swaps change the verdict. CHRONO-Fair extends this by (a) treating each flip as a survival event, which produces a time-resolved curve rather than a single number, (b) attaching the e-process to the sequential decision stream, which gives anytime-valid alarms, and (c) decomposing the flip mass into aleatoric and epistemic components. Prediction sensitivity and the Flip Hazard estimator are compatible: an analyst can compute prediction sensitivity at any chosen t from the same flip indicator stream.

7.3 Comparison to predictive multiplicity

The empirical Decision Flip Rate (eDFR) of Miller and Blume [77] is the closest reliability metric. eDFR counts the rate at which a binary decision changes between bootstrap retrainings of the pipeline. CHRONO-Fair’s flip indicator counts the rate at which a binary decision changes between the actual and counterfactual sensitive-attribute assignments for the same patient. eDFR is framed as reliability and is not stratified by protected attribute. CHRONO-Fair is framed as fairness and is stratified. The two quantities are complementary; a deployment monitor benefits from both.

7.4 Threats to internal validity

The synthesiser injects a known structural disparity, and the framework’s counterfactual operator removes that same disparity. Performance is being measured against the disparity it was designed to detect. To mitigate this, the false-alarm calibration experiment (Section 6.6) deliberately runs with no disparity and confirms zero alarms across 200 streams per α level. The decomposition Track 1 (Section 6.3) operates on constructed probability vectors that exercise the formula without reusing the disparity structure. Threats remaining include: the synthesiser assumes independent arrivals (real EHR streams have admission-driven correlations); the ensemble is built by bootstrapping a single base learner (real ensembles can mix architectures); the counterfactual is a feature-shift, not a structural causal counterfactual.

7.5 Threats to external validity

The streaming evidence in this paper is controlled rather than prospective, and that is its main external-validity threat. The real-data component is a Texas-100X batch verdict-stability analysis, while the sequential experiments run on Texas-100X-calibrated replay and simulation. A real admission feed differs from a calibrated stream in label delay, missingness, admission seasonality, and selection bias under triage, none of which the synthesiser reproduces in full. The conclusions should therefore be read as preliminary evidence about sequential monitoring behaviour, not as evidence of clinical effectiveness. Prospective validation on a live clinical stream is the necessary next step, and distribution-shift behaviour under performative effects [84] and dataset shift [38, 64] requires study beyond the present scope.

7.6 Statistical caveats

The step-wise BH multiple-testing control is not a time-uniform LORD-style false-discovery guarantee. The framework reports findings as “flagged at step t ”. Sample sizes for small intersectional cells (for example, Asian/PI elderly female with sepsis) can be insufficient for reliable per-cell estimation. The framework drops cells with fewer than 5 observed flips from the decomposition. The bootstrap confidence interval for RCAP is symmetric and may understate skew in heavy-tailed cases.

7.7 Clinical interpretation

A clinician inspecting the dashboard reads three signals: a survival curve (which patients are flipping and how fast), an e-process alarm (whether the change is statistically meaningful), and a diagnostic triage label (whether to escalate to a data audit or a retraining request). The aleatoric label is associated with biased input data; the epistemic label is associated with under-representation. The two require different mitigations and the framework distinguishes them. Schwartz et al. [95] and Tonekaboni et al. [102] report that clinicians prefer monitoring

outputs that translate into a concrete next step. The Inspector report is designed to satisfy that preference.

7.8 Regulatory positioning

The FDA PCCP final guidance [103] expects manufacturers to describe how the model will be monitored for bias after release. EU AI Act Articles 10, 14, and 17 [37] require continuous monitoring of high-risk medical AI under a documented bias-management plan. STANDING Together [3] provides consensus recommendations. The CHRONO-Fair framework provides a candidate technical building block for the monitoring layer of such a plan. Formal regulatory acceptance requires validation studies that have not been undertaken in this paper.

7.9 Scope statement

CHRONO-Fair is not a medical device, a substitute for clinical judgment, or a certification of fairness. It is a deployment-oriented prototype intended to expose fairness drift to clinical governance staff in time for intervention. Two empirical boundaries on this scope are stated explicitly. Operational adoption claims are deferred until prospective electronic-health-record stream validation has been completed on the deployment site’s own cohort. The AGE_GROUP monitoring behaviour reported here remains simulator-confirmed until a calendar-stamped Texas-100X release allows prospective record-order replay.

8 LIMITATIONS

The study has ten specific limitations. First, Texas-100X is the only empirical clinical dataset used. The findings have not been reproduced on a second cohort. Second, the streaming experiments are Texas-100X-calibrated controlled replay and simulation, not prospective live EHR deployment. Third, Texas-100X is administrative discharge data, not a real-time admission stream, so arrival order is reconstructed rather than observed. Fourth, the feature-shift counterfactual presumes a known marginal effect of the protected attribute on observable features. It is weaker than a structural causal counterfactual and under-attributes indirect effects, so the framework will tend to under-flag. Fifth, the e-process depends on a baseline flip rate ρ_0 . Section 6.13 shows that underestimating ρ_0 inflates the false-alarm rate, while overestimating it only delays detection. Sixth, small intersectional cells are unreliable. The detection rate falls below 0.95 for cells under roughly 1 000 monitored patients, and the empirical Wasserstein-1 RCAP estimate carries a small-sample upward bias. Seventh, label delay increases the wall-clock detection delay by approximately the delay length. Eighth, the step-wise Benjamini-Hochberg layer is a cross-sectional adjustment at each inspection time, not a time-uniform online false-discovery guarantee. Ninth, the dashboard and Inspector report are governance-support prototypes and have not been clinically validated with users. Tenth, CHRONO-Fair is a research prototype and not a medical device.

A further open question concerns performative effects [84]. If the framework’s alarms cause clinicians to change behaviour, the flip-rate process is no longer independent of past outputs, and the supermartingale guarantee holds only against a fixed pre-deployment baseline. Further data collection is required to determine exactly how performative feedback interacts with the e-process.

Two empirical gaps stated explicitly. Two gaps in the present evidence base remain open and bound the paper’s claims. First, the framework has not been evaluated on a prospective electronic-health-record stream. Operational adoption therefore requires prospective EHR-stream validation on the deployment site’s own admission cohort. The present paper makes

no operational-adoption claim until that validation is completed. Second, the AGE_GROUP cell is the residual unfair cell after pre-release adjustment. Its monitoring behaviour is currently confirmed only on the Texas-100X-calibrated streams, because the publicly released Texas-100 file does not carry calendar timestamps that would let it be replayed in arrival order. AGE_GROUP therefore remains simulator-confirmed pending prospective replay on a calendar-stamped Texas-100X release, or pending the equivalent timestamped release of the named-column file from the Texas Department of State Health Services. The capability table marks both gaps with “P” on the real-clinical-evidence axis (footnote of Table 2), and the present limitations paragraph is the place where those P-marks become an explicit boundary on what the manuscript claims.

9 CONCLUSION

CHRONO-Fair extends static VFR-style audit reliability into time-resolved monitoring of fairness-verdict flips. The real Texas-100X batch analysis shows that verdict instability is present in a real administrative cohort: one of 12 audited metric-attribute verdicts flips across hospital-network resamples. The Texas-100X-calibrated controlled streaming experiments show how the sequential monitor behaves under known drift, including its detection delay, its false-alarm control, its diagnostic decomposition, and its response to baseline miscalibration, small cells, and label delay. Together the two layers give time-resolved evidence about when and where a fairness verdict begins to flip, which a static batch audit cannot provide. The evidence here is batch-real and stream-controlled; it is not prospective clinical evidence. Prospective EHR-stream validation is the main next step, followed by a time-uniform online false-discovery procedure, a counterfactual-sensitivity study across the swap, feature-shift, and structural definitions, and a governance-user evaluation of the Inspector report.

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A REPRODUCIBILITY APPENDIX

Figure / Section	n	seeds	Drift configuration
Sec. 6.2, Fig. 3	10 000	0 to 49	onset $t = 5\,000$
Sec. 6.3 Track 1, Fig. 4	1 000	0 to 29	none
Sec. 6.4, Figs. 6, 7	15 000	42	none
Sec. 6.5, Fig. 9	20 000	11; bootstraps 0 to 999	none
Sec. 6.6, Fig. 10	10 000	0 to 199 per α	none (H_0)
Sec. 6.7, Fig. 11	12 000	4	onset $t = 6\,000$
Sec. 6.11, Fig. 14	12 000	0 to 29 per ρ_1	onset $t = 6\,000$; $\rho_1 \in [0.06, 0.20]$
Sec. 6.12, Fig. 15	15 000	42	none

Table 9. Per-experiment synthesiser configuration. All experiments are reproducible via `python -m chrono_fair.experiments.<expN>`.